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State of the Art on Transformer Pruning

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Abstract

The standard NLP architecture, transformers, is a successful approach to many different tasks. However, the size of the such model is costly to deploy. All parameters, heads, feed-forward neurons, and layers are stored and calculated in a dense model, even if some of these calculations are relatively meaningless for the final output given a specific task. The goal of pruning is to eliminate some of this computation but still allow for usable results. This report will investigate the definitions, scoring and application of pruning to transformers.

We begin with the simple formulation of pruning as a constrained compression problem and discuss the various criteria used to determine what can be removed, distinguish between unstructured sparsity and semi/structured pruning and different pruning regimes. In particular for Transformers, we are concerned with pruning what is possible to actually remove: the weights themselves, attention heads, feed-forward channels, or even entire layers.

However, one persistent drawback can be observed in the survey: the majority of techniques assign individual scores to the units of the model, before eliminating the lowest scoring ones. While this is straightforward, and often performs very well at moderate compression rates, it fails to properly address the more challenging problem of distributing a very small remaining budget over the entire model. At high compression ratios, the layer assignments can be crucial and are more important than the individual score of any unit. In conclusion, this deficiency is presented as the fundamental motivation behind search-based pruning methods.

Keywords: neural network pruning, Transformer compression, structured pruning, saliency criteria, search-based pruning

Résumé

Les modèles Transformer sont devenus des outils courants en traitement automatique du langage, mais leur taille rend leur déploiement coûteux. Un modèle dense stocke et exécute tous ses paramètres, toutes ses têtes d'attention, tous ses neurones feed-forward et toutes ses couches, même lorsqu'une partie de ce calcul contribue peu à la prédiction finale. L'élagage cherche à supprimer une partie de ce calcul tout en gardant un modèle exploitable. Ce rapport étudie la manière dont l'élagage a été formulé, évalué et appliqué aux modèles Transformer.

Le rapport commence par la formulation de l'élagage comme un problème de compression sous contrainte, puis passe en revue les principaux critères utilisés pour décider ce qui peut être retiré. Il présente les méthodes fondées sur la magnitude et les activations, les critères par gradient et développement de Taylor, ainsi que les approches du second ordre comme Optimal Brain Damage et Optimal Brain Surgeon. Il distingue aussi la parcimonie non structurée, semi-structurée et structurée, car ces choix n'ont pas les mêmes effets sur le matériel. Pour les Transformers, l'analyse se concentre sur les unités que l'on peut vraiment supprimer : les poids, les têtes d'attention, les canaux feed-forward et parfois des couches entières.

La revue fait apparaître une limite récurrente. Beaucoup de méthodes attribuent un score local aux unités, puis suppriment celles qui semblent les moins importantes. Cette stratégie reste utile à compression modérée, mais elle répond mal à une question plus difficile : comment répartir un budget très réduit sur l'ensemble du modèle ? À forte compression, la répartition finale entre les couches peut compter davantage que le score isolé d'une seule unité. Le rapport termine donc en présentant cette limite comme la principale motivation des méthodes d'élagage fondées sur la recherche.

Mots-clés : élagage de réseaux de neurones, compression de Transformers, élagage structuré, critères de saillance, élagage par recherche

ملخص

أصبحت نماذج المحوّلات أدوات قياسية في معالجة اللغات الطبيعية، إلا أن حجمها يجعل عملية نشرها باهظة التكلفة. فالنموذج الكثيف يخزن ويقيّم كل معلمة، وكل رأس انتباه، وكل خلية عصبية للتغذية الأمامية، وكل طبقة، حتى عندما تساهم بعض هذه العمليات الحسابية بقدر ضئيل في التنبؤ النهائي. لذلك يحاول التقليم إزالة جزء من هذه العمليات الحسابية مع الحفاظ على قابلية استخدام النموذج.

يدرس هذا التقرير التقليم في نماذج المحوّلات من حيث صياغته، ومعايير تقييمه، وطرائق تطبيقه. وينطلق من الصياغة الأساسية للتقليم بوصفه مسألة ضغط تخضع لقيود محددة، ثم يستعرض أهم المعايير التي تُستخدم لتحديد الأجزاء القابلة للحذف. وتشمل هذه المعايير الطرائق المعتمدة على مقدار الأوزان، والطرائق المعتمدة على التنشيطات، ومعايير التدرّج وتوسّع تايلور، إضافةً إلى المقاربات من الدرجة الثانية. ويميّز التقرير كذلك بين التقليم غير المنظّم، والتقليم شبه المنظّم، والتقليم المنظّم، لأن هذه الاختيارات تختلف في أثرها الفعلي على العتاد وسلوك التنفيذ. وبالنسبة إلى المحوّلات، يركز النقاش على الوحدات التي يمكن إزالتها عملياً: الأوزان، ورؤوس الانتباه، وقنوات التغذية الأمامية، وأحياناً طبقات كاملة.

تُظهر المراجعة قيداً متكرراً؛ إذ تقوم العديد من الطرق بتقييم الوحدات محلياً، ثم إزالة الوحدات ذات التصنيف الأقل. ويُعد هذا الأمر بسيطاً وفعالاً في الغالب عند مستويات الضغط المعتدلة، لكنه لا يجيب بشكل كامل عن السؤال الأصعب: كيف ينبغي توزيع ميزانية صغيرة متبقية عبر النموذج بأكمله؟ ففي مستويات الضغط العالية، يمكن أن يكون التخصيص النهائي عبر الطبقات أكثر أهمية من درجة أي وحدة منفردة. لذلك ينتهي التقرير بتأطير هذه الفجوة بوصفها دافعاً رئيسياً لطرق التقليم القائمة على البحث.

الكلمات المفتاحية: تقليم الشبكات العصبية، ضغط المحوّلات، التقليم المهيكل، معايير البروز والأهمية، التقليم القائم على البحث.

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List of Acronyms and Abbreviations

Adam	Adaptive Moment Estimation
AE	Autoencoder
BERT	Bidirectional Encoder Representations from Transformers
BF16	Brain Float 16
CE	Cross-Entropy Loss
CNN	Convolutional Neural Network
CoFi	Coarse-to-Fine Structured Pruning via Learned Masks
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DarwinLM	Training-Aware Evolutionary Structured Pruning
ELU	Exponential Linear Unit
EMA	Exponential Moving Average
EvoPress	Evolutionary Search over Layer-Wise Compression Profiles
FFN	Feed-Forward Network
FLOPs	Floating Point Operations
FP16	16-bit Floating Point
FP32	32-bit Floating Point
FT	Fine-Tuning
GA	Genetic Algorithm
GELU	Gaussian Error Linear Unit
GLUE	General Language Understanding Evaluation

GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
INT4	4-bit Integer
INT8	8-bit Integer
KD	Knowledge Distillation
KL	Kullback-Leibler Divergence
LLaMA	Large Language Model Meta AI
LLM	Large Language Model
LDLQ	LDL-Transpose Quantization
LLM-Pruner	Dependency-Aware Gradient-Based Structured Pruning
LoRA	Low-Rank Adaptation
MACs	Multiply-Accumulate Operations
MAE	Mean Absolute Error
MHA	Multi-Head Attention
MLP	Multi-Layer Perceptron
MNLI	Multi-Genre Natural Language Inference
MOP	Multi-Objective Optimization Problem
MRP	Multi-Weight Removal Process
MSE	Mean Squared Error
N:M	Semi-Structured Sparsity Pattern
NAS	Neural Architecture Search
NLP	Natural Language Processing
NSGA-II	Non-Dominated Sorting Genetic Algorithm II
OBS	Optimal Brain Surgeon
OPT	Open Pre-trained Transformer
PPL	Perplexity
QNLI	Question Natural Language Inference

ReLU	Rectified Linear Unit
RMSNorm	Root Mean Square Normalization
RNN	Recurrent Neural Network
SGD	Stochastic Gradient Descent
SparseGPT	Second-Order Unstructured Pruning via Hessian Compensation
SQuAD	Stanford Question Answering Dataset
SVD	Singular Value Decomposition
Tanh	Hyperbolic Tangent
Wanda	Weight and Activation-Based Pruning Criterion
ZipLM	Runtime-Aware Second-Order Structured Pruning

General Introduction

This rapid progress in NLP over the past several years can be mostly attributed to the Transformer architecture which is capable of learning arbitrarily long and complex dependencies between words in a parallelized fashion, and has already achieved state of the art accuracy for many possible sequence modeling tasks. The problem is that this expressive capability does not come free, as the high memory requirements and immense computational throughput and energy consumption associated with training these large models effectively raise the performance ceiling to a near impossible level for practical deployment.

The issue of closing the gap between the scale used for training and that for deployment has led to the proliferation of neural network pruning as a top compression technique. The ultimate goal of pruning is to detect and eliminate either redundant parameters or structures from the already trained or during the training of network and not alter its current performance much. Pruning is, in fact, not a single problem but a highly complex optimization. One has to consider not only the architecture of the pruning, but also the condition for pruning (Whether to rely on weights magnitudes, on gradients or loss curvature), and the condition needed to make recovery.

The growing literature on pruning Transformers has revealed a somewhat noticeable problem with current state-of-the-art which is that almost all existing techniques perform pruning via a local score. By this, we mean that the importance of an individual parameter or a group of them is computed independent of any other structural components, and without the ability to know the global effect of all pruning. While it works exceptionally well in low-compression situations, performance deteriorates dramatically when we desire high sparsity. At very aggressive sparsity levels, the correct distribution of the pruning budget over the model is far more important than the precision of the local scores.

The objective of this report is to review comprehensively the state-of-the-art in Transformer pruning, analyze the existing methodologies, and formally isolate this “allocation problem” to motivate future search-based pruning frameworks.

To achieve this, the report is divided into two main parts. The first part establishes the foundational knowledge required for this study and spans two chapters. Chapter 1 covers the theoretical foundations of neural networks. Chapter 2 then examines the architecture of the Transformer, defines its structural components and the specific hardware-aware efficiency metrics that practically drive the need for compression.

This second part of this report covers some of the current state-of-the-art literature of Transformer pruning. Chapter 3 introduces the formal definition of pruning in different structural

regimes. In Chapter 4 we analyze the existing work about Unstructured and Heterogeneous Search-based techniques for pruning. We mainly focus on methods that are trying to optimize flexibility and the space search at weight and layer profile level. In Chapter 5 we address the problem of Structured Methods where entire units are removed at block-level, so that the result is a hardware-oblivious dense-shape. Then, Chapter 6 gives a literature overview of the comparison of existing methods between different regimes and formalizes the main shortcomings of the current evaluation procedures by clearly pointing out the limitations of local scoring algorithms in finding the solution for the global allocation problem. Last but not least, in Chapter 7 we summarize our contribution and provide a well-defined motivation to treat future model compression as a global configuration search instead of a local rank.

Chapter 1

Foundations

1.1 Introduction

The purpose of pruning is to keep the behavior of a trained neural network while modifying its size by removing some of its parameters or biggest parts. In order to study what is eligible to be removed and how the network will be effected by removing these parts it is first needed to explain what are the objects that pruning can work on: Neurons, weights, activations, layers, feed-forward neural network. The vocabulary is not given here as general knowledge.

We begin this chapter by giving a description of a single artificial neuron. This is necessary for pruning because weights and biases are the smallest parameters which are available for training and activation determines where the information should propagate to after weights and biases have been updated. It naturally leads us to multi-layer perceptrons, loss functions, gradient based methods and backpropagation because we measure the performance of a pruning technique by the amount to which the output behavior, or the loss, changes when removing some part of a network[Goodfellow et al., 2016; Rumelhart et al., 1986].

1.2 Parameterization of Neural Networks

An artificial neuron is a computing unit that takes an input vector, it applies a linear transformation and finally inputs this result to a non-linear activation function. This form of an artificial neuron enables the artificial neural network to learn non-linear mappings. The linear part is also called an affine transformation, which computes a weighted sum of the input vector and then adds a bias. The non-linear activation function is important to enable the artificial neural network to learn complex patterns, since it is responsible for approximate the non-linearities in the data. Examples for common activation functions are ReLU, sigmoid and tanh.

1.2.1 The Neuron as a Linear Model

The artificial neuron, formally introduced by [McCulloch & Pitts, 1943], serves as the fundamental building block of neural networks. Mathematically, it performs an **affine transformation** on its inputs followed by a non-linear activation f which is represented in Figure 1.1 and expressed in Eq.(1.1).

$$z = \boldsymbol{w}^T \boldsymbol{x} + b = \sum_{i=1}^{n} w_i x_i + b \tag{1.1}$$

Given an input $\boldsymbol{x} \in \mathbb{R}^n$, weights $\boldsymbol{w} \in \mathbb{R}^n$ scale the importance of each feature, while the bias $b \in \mathbb{R}$ acts as an offset. Figure 1.1 shows this affine unit before the activation is applied: the weighted inputs are summed with the bias to produce the pre-activation response z.

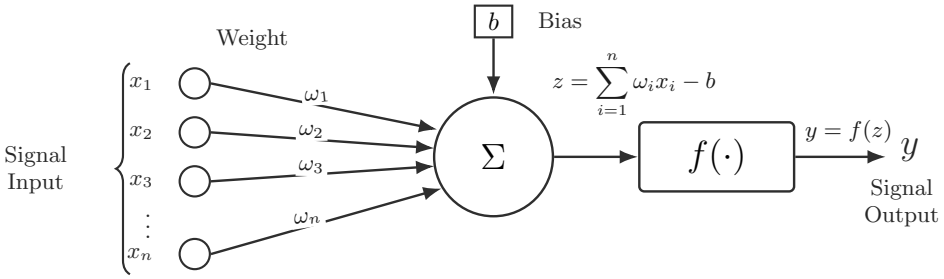


Figure 1.1: A single artificial neuron with input vector $\boldsymbol{x}$, weight vector $\boldsymbol{w}$, bias b, and pre-activation output z. The activation function f is applied to z to produce the final output y.

The term $\boldsymbol{w}^T \boldsymbol{x}$ measures how strongly the input matches the learned weights. This affine computation is the basic operation repeated across layers in larger neural networks.

1.2.2 Activation Functions

The neural network's ability to model complex patterns depends on the non-linear activation functions applied after each affine transformation. If we were to stack multiple layers without any non-linearity, the entire network would collapse into a single affine transformation.

Therefore, depth alone is not enough: neural networks require non-linear **activation functions** to model complex patterns [Goodfellow et al., 2016].

As shown in Figure 1.1, the activation function f takes the pre-activation output z and produces the final output $y = f(z)$. The choice of activation function can significantly affect the network's performance, convergence, and ability to learn from data.

Classical Saturating Activations

Sigmoid (Eq.(1.2)) and hyperbolic tangent (Eq.(1.3)) functions were often used in early networks. While these functions take real-valued inputs and output bounded values, which could be useful in the case of probabilities or normalized hidden states, they have very small derivatives when the input is very large or very small. This saturation causes learning to be very slow in deep networks [Glorot & Bengio, 2010; Goodfellow et al., 2016].

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \qquad \sigma : \mathbb{R} \to (0, 1). \tag{1.2}$$

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}, \qquad \tanh : \mathbb{R} \to (-1, 1). \tag{1.3}$$

Rectified Activations

Rectified Linear Units (ReLU) became common in deep learning because they are simple, computationally cheap, and reduce saturation for positive inputs [Nair & Hinton, 2010]. ReLU is defined as showing in Eq.(1.4):

$$\mathrm{ReLU}(x) = \max(0, x), \qquad \mathrm{ReLU} : \mathbb{R} \to [0, \infty). \tag{1.4}$$

Leaky ReLU keeps a small non-zero slope for negative inputs, reducing the risk that a unit becomes permanently inactive [Maas et al., 2013], as shown in Eq.(1.5):

$$\mathrm{LeakyReLU}(x) = \begin{cases} x, & x \geq 0, \\ \alpha x, & x < 0, \end{cases} \qquad \alpha \in (0, 1), \quad \mathrm{LeakyReLU} : \mathbb{R} \to \mathbb{R}. \tag{1.5}$$

The Exponential Linear Unit (ELU) uses a smooth negative branch and a linear positive branch [Clevert et al., 2016], which can lead to faster convergence and improved performance in some cases, as shown in Eq.(1.6):

$$\mathrm{ELU}(x) = \begin{cases} x, & x \geq 0, \\ \alpha(e^{x} - 1), & x < 0, \end{cases} \qquad \alpha > 0, \quad \mathrm{ELU} : \mathbb{R} \to (-\alpha, \infty). \tag{1.6}$$

Modern Transformer architectures often use the Gaussian Error Linear Unit (GELU), a smooth activation that weights inputs by their value under the standard Gaussian cumulative

distribution [Hendrycks & Gimpel, 2016], as shown in Eq.(1.7):

$$\mathrm{GELU}(x) = x\Phi(x) \approx \frac{x}{2}\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}(x + 0.044715x^3)\right)\right). \tag{1.7}$$

Figure 1.2 compares the shapes of these activation functions on the same input range. The plot makes the difference between saturating functions and rectified functions visible.

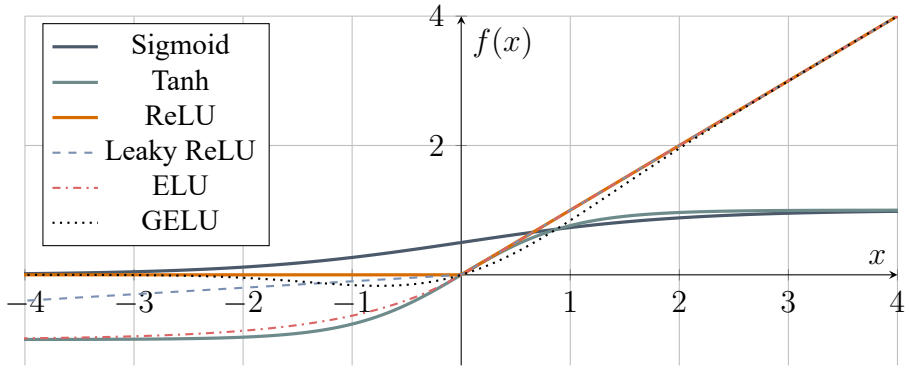


Figure 1.2: Comparison of common activation functions

1.2.3 The Multi-Layer Perceptron (MLP)

Although a single active neuron can map an input to a basic response, any practical model would require many neurons to co-operate together. In a Multi-Layer Perceptron, neurons are stacked into layers where one layer transforms the output of the preceding one.

Going from a single neuron to an MLP increases expressiveness by composing affine transformations with non-linear activations across layers [Rumelhart et al., 1986]. MLPs are *feed-forward* and typically *fully connected*, meaning each neuron connects to all neurons in the next layer.

Eq. (1.8) formalizes the computation performed by each layer of an MLP, to transform the raw input $\boldsymbol{x}$ into the final output (prediction)$\boldsymbol{y}$.

$$\begin{aligned} \boldsymbol{h}^{(0)} &= \boldsymbol{x}, \\ \boldsymbol{h}^{(\ell)} &= f(\boldsymbol{W}^{(\ell)}\boldsymbol{h}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}) \quad \text{for } \ell = 1, 2, \ldots, L-1, \\ \boldsymbol{y} &= f_{\text{out}}(\boldsymbol{W}^{(L)}\boldsymbol{h}^{(L-1)} + \boldsymbol{b}^{(L)}), \end{aligned} \tag{1.8}$$

Figure 1.3 illustrates a typical MLP with one input layer, two hidden layers, and one output layer.

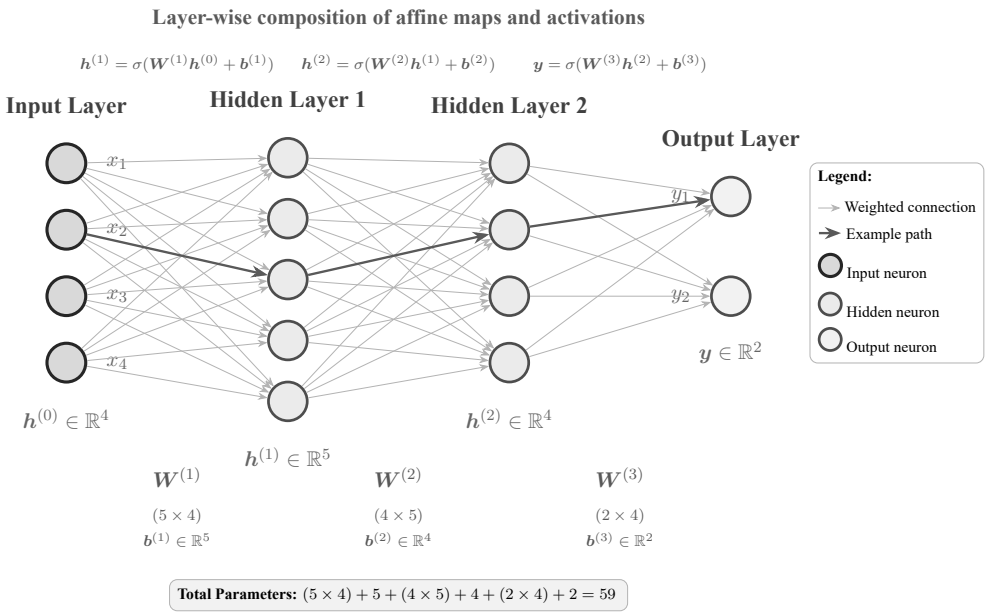


Figure 1.3: Fully connected MLP with 4 input features, two hidden layers (5 and 4 neurons), and 2 output neurons.

1.3 Optimization and Gradients

We outlined how an MLP generates a prediction which is by a composition of transformations applied at each layer. With forward computation, learning can be thought of as figuring out what the weights and biases should be given the data. Network construction then becomes an optimization problem: select the parameters that make the output as close as possible to the target on a training set.

Hence, a quantitative measure must be provided that can actually be minimized; this provides the rationale for empirical risk minimization.

1.3.1 Empirical Risk Minimization

Given a dataset $\mathcal{D} = \{(\boldsymbol{x}^{(i)}, y^{(i)})\}_{i=1}^{m}$, learning consists of finding parameters $\boldsymbol{\theta}$ that minimize the discrepancy between the model predictions $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ and the target values. This is commonly formulated as empirical risk minimization [Vapnik, 1998]:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \hat{R}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}), y^{(i)}). \tag{1.9}$$

where m is the number of training samples; $\boldsymbol{x}^{(i)}$ and $y^{(i)}$ are the input and target of the i-th sample; $\boldsymbol{\theta}$ are the model parameters; $\hat{R}(\boldsymbol{\theta})$ is the empirical risk, i.e. the average loss over the training set; and $\mathcal{L}(\cdot, \cdot)$ is the per-sample loss function. For a K-class classification problem the prediction is a probability vector $\hat{\boldsymbol{y}} \in \Delta^K$ with entries $\hat{y}_k$, the target $\boldsymbol{y}$ has entries y_k, and c denotes the index of the true class; the Kullback–Leibler divergence [Cover & Thomas, 2006] $D_{\mathrm{KL}}(\boldsymbol{y} \,\|\, \hat{\boldsymbol{y}})$ measures the excess cost of using $\hat{\boldsymbol{y}}$ in place of $\boldsymbol{y}$.

This empirical-risk objective in Eq. (1.9) gives the general learning template: choose parameters by minimizing the average loss over the training set. It does not specify the loss itself. That choice is determined by the problem and the structure of the output.

Loss Functions

Now we have an optimization problem. But what are we minimizing? A loss function accounts for the discrepancy between predictions and targets and makes explicit assumptions about the problem and the output space. Table 1.1 summarizes the most common loss functions used in neural network training, along with their formulas, target types, and properties. The choice of loss function depends on the task at hand (classification vs regression) and the nature of the targets (hard one-hot labels, soft probability distributions, or continuous real values.).

Table 1.1: Summary of common neural network loss functions.

Loss	Task	Formula	Target type	properties
Cross-Entropy	Classification	$\mathcal{L}_{\text{CE}}(\hat{\boldsymbol{y}}, \boldsymbol{y}) = -\sum_{k=1}^{K} y_k \log \hat{y}_k$	Soft or one-hot	Measures total encoding cost of $\boldsymbol{y}$ under code optimal for $\hat{\boldsymbol{y}}$; reduces to $-\log \hat{y}_c$ for hard labels.
KL Divergence	Classification	$D_{\text{KL}}(\boldsymbol{y} \,\|\, \hat{\boldsymbol{y}}) = \sum_{k=1}^{K} y_k \log \frac{y_k}{\hat{y}_k}$	Soft distribution	Measures *excess* encoding cost; $D_{\text{KL}} = \mathcal{L}_{\text{CE}} - H(\boldsymbol{y})$. Preferred over CE when targets are soft.
MSE (L2)	Regression	$\mathcal{L}_{\text{MSE}}(\hat{y}, y) = \frac{1}{m}\sum_{i=1}^{m}(\hat{y}^{(i)} - y^{(i)})^2$	Continuous	Penalises errors quadratically; equivalent to maximum likelihood estimation (MLE) under Gaussian noise. Sensitive to outliers.
MAE (L1)	Regression	$\mathcal{L}_{\text{MAE}}(\hat{y}, y) = \frac{1}{m}\sum_{i=1}^{m}\lvert\hat{y}^{(i)} - y^{(i)}\rvert$	Continuous	Penalises errors linearly; robust to heavy-tailed distributions and outliers.

Notes. K: number of classes; $\hat{\boldsymbol{y}} \in \Delta^K$: predicted probability vector (softmax output); $\boldsymbol{y}$: target distribution (one-hot or soft); c: index of the true class; $H(\boldsymbol{y}) = -\sum_k y_k \log y_k$: entropy of the target; m: number of samples; $\hat{y}^{(i)}, y^{(i)}$: predicted and true scalar outputs for sample i. When $\boldsymbol{y}$ is one-hot, $H(\boldsymbol{y}) = 0$ and $D_{\text{KL}} = \mathcal{L}_{\text{CE}}$.

1.3.2 Algorithms for Gradient-Based Optimization

With the loss function established, the remaining question is how the resulting error signal propagates back through the network to update the parameters.

Training neural networks is usually a problem of finding a parameter vector $\theta \in \mathbb{R}^d$ that minimizes the scalar objective $J(\theta)$ where J is the empirical risk defined by the loss function and the training data. Formally, we want to solve (1.10):

$$\theta^* = \arg\min_{\theta} J(\theta). \tag{1.10}$$

The parameter space in most contemporary neural networks is high-dimensional and the objective is largely non-convex so a solution in closed form for optimum parameters rarely

exists. Training is hence iterative: the parameters are updated according to directions suggested by the gradient of the loss [Bottou et al., 2018].

The gradient $\nabla J(\theta)$ indicates the direction of steepest local increase of the objective, and optimizers will use it in reverse to adjust the parameters to yield a lower loss. This concept forms the basis of the gradient descent algorithm as well as of all the adaptive optimization schemes currently used in training neural networks.

Gradient Descent

Gradient descent updates parameters in the negative-gradient direction. For iteration t and learning rate $\eta > 0$, the update is formulated in Eq.(1.11):

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t). \tag{1.11}$$

The learning rate controls the step size. If it is too small, optimization may progress slowly; if it is too large, the updates can overshoot useful regions of the objective surface. In large scale learning, the exact gradient over the full dataset is often replaced by a stochastic estimate computed on a mini-batch. This gives stochastic gradient descent, which is cheaper per update and is widely used for neural-network training [Bottou et al., 2018].

Adam

Adam is an adaptive gradient-based optimizer that combines momentum with per-parameter learning-rate adaptation [Kingma & Ba, 2015]. Instead of using only the current gradient, Adam keeps exponential moving averages of the first and second moments of the gradient, which are updated as shown in Eq.(1.12):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \qquad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \tag{1.12}$$

where $g_t = \nabla J(\theta_t)$ is the gradient estimate at iteration t, m_t tracks the average direction of recent gradients, and v_t tracks their squared magnitude.

Because both moving averages are initialized at zero, Adam applies bias correction computed as follows Eq.(1.13):

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}. \tag{1.13}$$

The parameter update is then given by (1.14):

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \tag{1.14}$$

Adam is often utilized due to its robustness in relation to scaling differences among gradients for different parameters and typically requires less fine-tuning of learning rates as opposed

to simple gradient descent. Subsequent versions of Adam include AdamW, which removes the weight decay from the adaptive learning component and is applicable to a variety of deep learning applications [Loshchilov & Hutter, 2019].

Both gradient descent and Adam describe how the parameters should be updated, but they both require that the gradients be available. These computations can become expensive in deep, multi-layered networks and thus require an alternative method, termed backpropagation.

1.3.3 Backpropagation

Backpropagation computes these gradients efficiently by applying the chain rule from the output layer back through the network [Rumelhart et al., 1986]. In compact form, it provides the quantities $\nabla_\theta \mathcal{L}$ where θ denotes the trainable parameters and $\mathcal{L}$ is the loss function.

In the case of this pruning-oriented report we do not require the full derivation above. It is sufficient that back-propagation relates model error to specific parameters and structures, which will be exploited by the gradient-based pruning criteria, Taylor approximation, Fisher scores and post-pruning recovery process detailed in later chapters.

1.4 Conclusion

In this chapter, we have established the mathematical and optimization framework of deep neural networks. We described the building blocks-the affine and non-linear transformation of feed forward networks and formulated training of a feed forward network as an empirical risk minimization problem. We also described the central optimization concepts of gradient and Hessian matrices, crucial to the learning process of the networks and to pruning algorithms in assessing parameters relevance.

Though the above theory applies to any feed-forward network, the architecture used in modern NLP are often extremely specialized. To know how pruning is practically applied one must be aware of the mapping of generic matrix operations to the structures within an architecture. Therefore the subsequent chapter deals solely with the Transformer architecture and its components, how parameters are structured and the drastic bottleneck which necessitates compression.

Chapter 2

Transformer Architectures and Efficiency

2.1 Introduction

While Multi-Layer Perceptrons offer a generic learning model over vector inputs, they do not inherently leverage any underlying structure of the data. For this report, the architecture of interest is the Transformer [Vaswani et al., 2017]. Since its introduction, the Transformer's ability to model long-range dependencies in parallel has made it the foundation of state-of-the-art sequence modeling. However, this expressive power comes at a severe computational cost. To effectively compress such a model, one cannot treat it as a generic sequence of matrices; hence, one must deeply understand its internal wiring, where its parameters are concentrated, and how data moves through its layers during execution.

The objective of this chapter is to investigate the Transformer architecture using computational costs and parameterization of the structure. Based on this objective, it describes the key modules within the network, in particular the Multi-Head Attention and the position-wise FFN, and compares their functional roles with respect to the parameters to which the subsequent compression techniques will refer.

These components provide the building blocks for a discussion of the overall Encoder-Decoder architecture in this chapter before moving onto the major drivers of this report; the computational bottleneck; examining the quadratic scaling of self-attention and the very large memory demands for dense weight matrices. This chapter concludes by mentioning the relevant hardware-aware efficiency measures (FLOPs, memory bandwidth and latency).

2.2 Transformer Networks

The Transformer network, introduced in [Vaswani et al., 2017], shifted sequence modeling by processing with attention rather than recurrence. Unlike the earlier recurrent models that processed sequence words sequentially, the Transformer can simultaneously consider all sequence words. This makes the model effective for language modeling, machine translation, summarization, and numerous other sequence based problems. The first Transformer was an encoder-decoder architecture for machine translation, though in practice, encoder-only architectures like BERT [Devlin et al., 2019] and decoder-only architectures like LLaMA [Touvron et al., 2023] are more common.

2.3 Core Architectural Components: Building Blocks of the Transformer

It would be beneficial to list the components present in the transformer layers (before dividing the model into encoder and decoder stacks), since those are the building blocks of the repeated components. These layers consist of the following building blocks: token embeddings, in which a token represents a single, distinct unit (a word, word piece or punctuation) in the input sequence after text preprocessing; positional information, attention projections, feed forward channels, residual connections and normalization layers. The primary components to compress are those listed previously, which also largely define the model's parameter count and computational cost.

Input Representation: Embeddings and Positional Encoding

Every token is mapped to a continuous embedding vector. Since self-attention does not inherently encode token order, positional information is added to the token representations. The original Transformer uses sinusoidal positional encodings [Vaswani et al., 2017], shown in Eq.(2.1), where pos is the position of the token in the sequence, i is the index of the embedding-dimension pair, and d_{model} is the embedding size:

$$PE_{(\text{pos},2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right), \quad PE_{(\text{pos},2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right). \tag{2.1}$$

Self-Attention Mechanism

Self-attention allows each token representation to aggregate information from the rest of the sequence. The mechanism uses three learned projections of each input: queries, keys, and values [Vaswani et al., 2017]. Queries represent what a token is looking for, keys represent what each token offers for comparison, and values contain the information that will be aggregated.

For an input sequence representation $\boldsymbol{X} \in \mathbb{R}^{n \times d_{\text{model}}}$, the projections are computed as shown in Eq.(2.2), where $\boldsymbol{W}^Q, \boldsymbol{W}^K, \boldsymbol{W}^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are learned weight matrices for queries, keys, and values respectively:

$$\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}^Q, \quad \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}^K, \quad \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}^V. \tag{2.2}$$

Scaled dot-product attention is then computed as shown in Eq.(2.3), where d_k is the dimension of the queries and keys:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V}. \tag{2.3}$$

The factor $\sqrt{d_k}$ prevents dot products from becoming too large, which would push the softmax into saturated regions with weak gradients.

Multi-Head Attention

Multi-head attention applies several attention mechanisms in parallel, allowing the model to attend to different representation subspaces and token relationships at the same time [Vaswani et al., 2017]. Eq.(2.4) defines the computation for each head i, where $\boldsymbol{W}_i^Q, \boldsymbol{W}_i^K, \boldsymbol{W}_i^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are the projection matrices for head i:

$$\text{head}_i = \text{Attention}(\boldsymbol{X}\boldsymbol{W}_i^Q, \boldsymbol{X}\boldsymbol{W}_i^K, \boldsymbol{X}\boldsymbol{W}_i^V), \quad i = 1, \ldots, h. \tag{2.4}$$

The head outputs are concatenated and projected back to the model dimension as expressed in Eq.(2.5), where $\boldsymbol{W}^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ is the output projection matrix:

$$\text{MultiHead}(\boldsymbol{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\boldsymbol{W}^O. \tag{2.5}$$

Feed-Forward Network

After attention mixes information across tokens, a position-wise Feed-Forward Network (FFN) processes each token representation independently The FFN usually expands the hidden representation to a larger intermediate dimension d_{ff} and then projects it back to d_{model}. In the original Transformer, d_{ff} is typically much larger than d_{model}, making FFN channels a major source of parameters and a natural target for structured pruning.

Residual Connections and Layer Normalization

Each attention or FFN sub-layer is wrapped by a residual path and layer normalization. Residual connections preserve a direct path for information and gradients, while normalization stabilizes the scale of hidden representations. This design is essential for training deep Transformer stacks and is also important when later compression methods remove heads, channels, or blocks [Ba et al., 2016; He et al., 2016; Vaswani et al., 2017].

2.3.1 Encoder and Decoder Architecture

With the core components defined, the encoder and decoder can be described cleanly. The full encoder-decoder Transformer contains two main stacks:

- **Encoder:** Processes the complete input sequence in parallel and produces contextual source representations.
- **Decoder:** Generates the output sequence one token at a time from left to right (autoregressively), attending to both previously generated target tokens and the encoder output.

The encoder converts the source sequence into a contextual memory representation $\boldsymbol{H}_{\text{enc}}$. The decoder uses this memory through cross-attention while also using masked self-attention to prevent access to future target tokens. The complete architecture is shown in Figure 2.1.

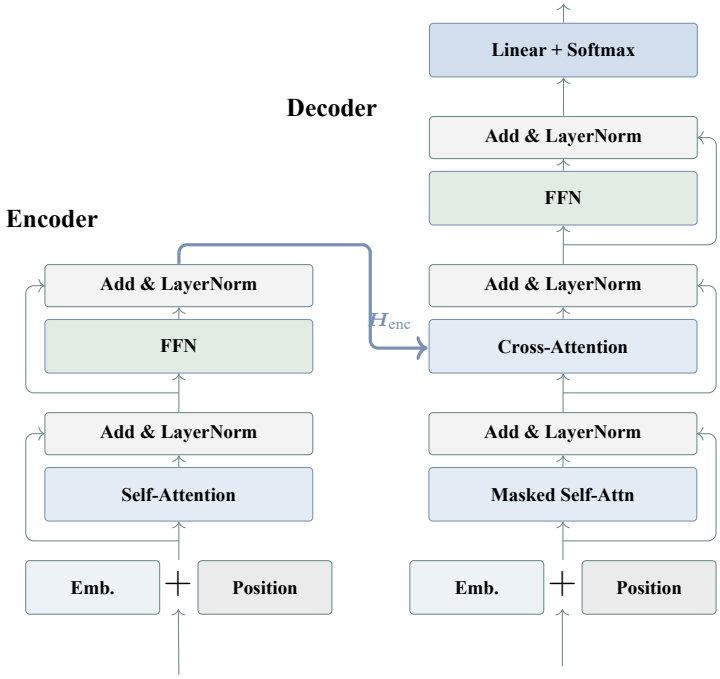


Figure 2.1: Complete Encoder-Decoder Transformer architecture

The Encoder Stack

An encoder layer contains multi-head self-attention followed by a feed-forward network, with residual connections and normalization around each sub-layer. In encoder self-attention, the queries, keys, and values all come from the same source sequence. This makes the encoder bidirectional: every input token can attend to every other input token. Figure 2.2 shows the internal order of these operations before the resulting representation is passed to the decoder. After N encoder layers, the output is the memory matrix $\boldsymbol{H}_{\text{enc}} \in \mathbb{R}^{n \times d_{\text{model}}}$, which summarizes the source sequence for the decoder.

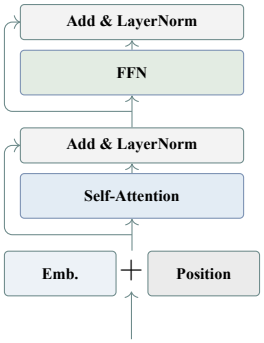


Figure 2.2: Transformer encoder layer architecture

The Decoder Stack

A decoder layer adds masked self-attention and cross-attention before the feed-forward network, enabling autoregressive generation while attending to encoder outputs. Masked self-attention restricts token t to positions $\leq t$, preventing information leakage from future target tokens. Cross-attention then links the decoder to the encoder: decoder states act as queries, while encoder states provide keys and values. Figure 2.3 shows this additional cross-attention block, which is the main structural difference between the decoder layer and the encoder layer.

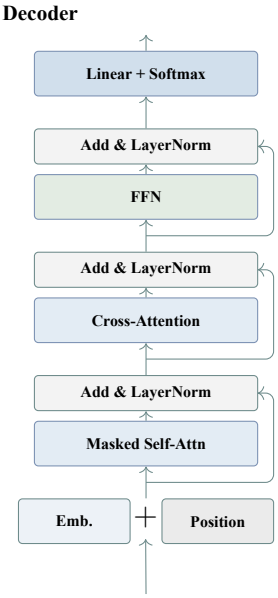


Figure 2.3: Transformer decoder layer architecture

Encoder-Decoder Information Flow

The complete sequence-to-sequence computation proceeds as follows:

1. **Encoding phase.** Source tokens $\boldsymbol{x} = (x_1, \ldots, x_n)$ are embedded, combined with positional information, and passed through N encoder layers. The output is $\boldsymbol{H}_{\text{enc}}$.

2. **Decoding phase.** Target tokens $\boldsymbol{y} = (y_1, \ldots, y_t)$ are embedded and processed one token at a time from left to right (autoregressively), so that each token is predicted only from the tokens before it. Each decoder layer applies masked self-attention, cross-attention over $\boldsymbol{H}_{\text{enc}}$, and a feed-forward transformation.

3. **Output phase.** The final decoder representation is projected to vocabulary logits. A softmax layer converts logits into token probabilities, and the selected token is fed back into the decoder at the next step.

At this point in the report, we can view the Transformer as an arrangement of embedding layers, attention layers, feed-forward functions, and encoder–decoder connections. This architecture accounts for why it performs well on sequence modelling tasks, since each position can refine its representation using information from the rest of the sequence. This expressive structure, however, is computationally demanding. The next point examines where this cost arises inside the Transformer architecture.

2.3.2 The Computational Bottleneck of Transformers

While powerful, the Transformer's success comes at a significant computational cost, which forms a central motivation for the compression techniques explored in this report. The standard Transformer architecture replaces recurrence with self-attention, enabling strong parallelism and high performance across sequence modelling tasks [Vaswani et al., 2017]. However, this advantage comes with a major bottleneck: in full self-attention, each token attends to every other token in the sequence. For a sequence of length n, this requires an attention score matrix of size $n \times n$, making the attention component scale quadratically with sequence length in both computation and memory [Beltagy et al., 2020; Tay et al., 2022].

This quadratic behaviour has direct practical implications. Doubling the sequence length from 512 to 1024 multiplies the attention matrix size by four, since $1024^2/512^2 = 4$. As a result, processing long documents or long-context inputs becomes substantially more expensive, which has motivated a large body of work on efficient Transformer variants and sparse or linear attention mechanisms [Beltagy et al., 2020; Tay et al., 2022].

At the same time, modern language models benefit strongly from scaling model size, training data, and compute [Kaplan et al., 2020]. Later scaling-law work further showed that model size and the number of training tokens should be scaled together under a fixed compute budget, highlighting the continuing importance of large-scale training [Hoffmann et al., 2022]. This creates a tension between scale and deployment: larger models often achieve better performance, but practical inference is constrained by latency, memory, energy, and hardware availability. This tension motivates compression methods such as pruning, quantization, and knowledge distillation [Han et al., 2016; Sanh et al., 2019].

The computational bottleneck defined above can be considered in several different ways. A Transformer may be expensive because of the size of its parameters, the memory required for activations, the number of arithmetic operations it performs, or the time required to run it on a specific hardware platform. Before discussing model efficiency in more detail, it is therefore

useful to introduce the most common measures used to assess computational cost.

2.3.3 Hardware-Aware Efficiency Metrics

Compression is evaluated through several distinct efficiency metrics. Parameter count, memory footprint, arithmetic cost, latency, and throughput capture different aspects of model execution. Hardware-aware evaluation therefore requires separate treatment of storage cost, arithmetic cost, and runtime behavior [Sze et al., 2017].

Parameter Count and Model Size

Let $\boldsymbol{\theta} = \{W^{(l)}, b^{(l)}\}_{l=1}^{L}$ denote the parameters of a network. The total parameter count is definded by Eq.(2.6):

$$P = \sum_{l=1}^{L} \left(|W^{(l)}| + |b^{(l)}| \right). \tag{2.6}$$

If the parameters are stored with precision b_w bits. Eq.(2.7) gives the model size in bytes:

$$M_{\text{model}} = \frac{P\, b_w}{8} \quad \text{bytes.} \tag{2.7}$$

This follows from counting the stored elements and multiplying by the number of bytes used by each element [PyTorch Contributors, 2026a, 2026b]. This quantity determines whether a model fits within device memory and how much parameter traffic must be sustained during inference. Quantization reduces b_w, pruning reduces the number of active parameters, and low-rank factorization reduces the parameterization itself.

MACs, FLOPs, and Arithmetic Intensity

For a dense matrix multiplication $Y = AB$ with $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$, the arithmetic cost is defined by Eq. (2.8):

$$\text{MACs}(A, B) = mkn, \qquad \text{FLOPs}(A, B) \approx 2mkn. \tag{2.8}$$

One multiply-accumulate corresponds to one MAC, while FLOP counting often treats the multiply and the addition as two floating-point operations [NVIDIA, 2026]. In Transformer layers, the feed-forward subnetwork contributes large dense matrix products, while self-attention adds the sequence-dependent $O(n^2 d_{\text{model}})$ cost of score computation and value aggregation.

Latency, and Throughput

Latency and throughput characterize runtime from complementary viewpoints. Latency is the time required to complete one inference request or one decoding step [Sze et al., 2017] as shown in Eq.(2.9):

$$T_{\text{latency}} = T_{\text{compute}} + T_{\text{memory}} + T_{\text{overhead}}. \tag{2.9}$$

Throughput measures completed work per unit time, for example Eq.(2.10) gives requests per second or generated tokens per second [Sze et al., 2017]:

$$\text{Throughput} = \frac{N_{\text{outputs}}}{T}. \tag{2.10}$$

2.4 Conclusion

In this chapter, we discussed in detail the structure of Transformer architecture and its computation analysis. After breaking down its major modules, such as Multi-Head Attention and Feed-Forward Networks, we pin-pointed exactly where these large majority parameters of this model are, and discovered the computation bottle necks this architecture imposes, that is, the quadratic complexity of self-attention mechanism and extremely huge bandwidth requirement for streaming the dense weight matrices in inference.

With these structures established, the hardware-aware metrics help relate them to deployment cost. Changes in parameter count must directly influence memory, FLOPs, arithmetic intensity, or latency to have an effect on deployment cost. From these, with our foundation from Chapter 1, we can proceed with modification to the weights and structure by introducing Neural Network Pruning as a systematic methodology for identifying and removing non-critical weights and structures to achieve efficient deployment.

Chapter 3

Neural Network Pruning

3.1 Introduction

Within the framework established in Chapters 1 and 2, pruning can be understood as a constrained reduction of a trained network. The optimization perspective defines how pruning error is measured, the architectural structure of the network determines which components can be removed, and deployment metrics determine whether pruning leads only to nominal sparsity or to practical savings.

Pruning has been a well-studied method for deep neural network compression. The objective of it is to eliminate non-essential active parameters or operations of a network without substantially changing the behavior of the pruned network as compared to a reference network. When the elements are removed from the network, the real topology of the network is altered either by setting (masking) the weights, channels, filters, layers, or other units to zeros or by completely removing the elements. The masking choice is both an architectural decision and a compression decision.

The relevance of pruning follows from the overparameterised nature of modern neural networks where large models often contain substantial redundancy distributed unevenly across layers and parameter groups [Denil et al., 2013; Michel et al., 2019]. Table 3.1 illustrates the scale: each parameter requires 4 bytes at full precision (FP32) or 2 bytes at half precision (FP16), placing a hard memory floor on deployment before activations or caches are considered [Devlin et al., 2019; He et al., 2016; OpenAI et al., 2023; Radford et al., 2019; Touvron et al., 2023].

Table 3.1: Parameter counts and memory footprints for selected modern models. FP32 assumes 4 bytes per parameter; FP16 assumes 2 bytes per parameter.

Model	Params	FP32 (GB)	FP16 (GB)	Notes
ResNet-50	25.6 M	0.10	0.05	Image classification
BERT-Base	110 M	0.44	0.22	Encoder model
GPT-2 (XL)	1.56 B	6.24	3.12	Left-to-right text generator
LLaMA-2 (7B)	6.74 B	26.9	13.5	Large decoder model
LLaMA-2 (70B)	68.9 B	275	138	Requires multiple GPUs
GPT-4	–	–	–	Parameters not publicly disclosed

Sparsity is simply not the right condition to prune on. We cannot expect to decrease for example the 140GB full precision footprint for a 70B-parameter model if we just zero out half the weights, as practical acceleration is only achieved if the sparse pattern can be leveraged by the software and hardware stack[Cheng et al., 2024a; Gale et al., 2020]. The point of pruning is then to decide what components of the network will be kept active for a given budget.

3.2 Theoretical Framework of Pruning

The process of pruning can be stated as a constrained optimization problem, namely to find a binary mask that specifies the active parameters and minimally increases the loss. Once this mask is selected, applying it to the weight tensors will produce a pruned network. From this formulation, the optimization problem could be viewed as the explicit constrained problem, where we are limited in the number of active parameters, or as a regularized objective, where we push the network towards sparsity by including a penalty term in the loss. For this reason, since finding the exact optimal mask is NP-hard, pruning methods make use of heuristics for importance, where importance is typically defined via a Taylor expansion of the loss function to estimate the change in the objective.

3.2.1 Formal Problem Formulation

Given a parameterized network $f(\cdot;\Theta)$ whose P parameters are stored in $\Theta \in \mathbb{R}^P$, a binary mask $M \in \{0,1\}^P$ is chosen such that those parameters selected to be kept (whose mask value is 1) can be retrieved and those not (mask value 0) excluded from further calculation by computing $f(x; M \odot \Theta)$, which zeros out parameters for which $M_i = 0$ by elementwise product.

The goal is then to find the mask M and the remaining weights Θ that keep the loss $\mathcal{L}$ as low as possible while meeting a budget on how many parameters are retained:

$$\min_{\Theta, M} \; \mathcal{L}(M \odot \Theta) \quad \text{subject to} \quad \frac{\|M\|_0}{P} \leq \kappa, \tag{3.1}$$

where $\|M\|_0$ counts the nonzero entries of M and $\kappa \in (0,1]$ is the target density. In practice this hard constraint is often replaced by a regularised objective that avoids having to fix κ in advance:

$$\min_{\Theta, M} \; \mathcal{L}(M \odot \Theta) + \lambda\, \Omega(M), \tag{3.2}$$

where λ controls how aggressively compression is penalised and $\Omega(M)$ can capture parameter count, memory footprint, or latency depending on the deployment target. This regularised form folds the budget into the objective instead of enforcing it as a hard constraint (Eq. (3.2)).

Finding an optimal pruning solution is generally NP-hard, since the search space grows exponentially with the number of parameters [Dong et al., 2017]. Practical methods thus resort to approximate measures which evaluate how pruning a parameter or structure affects performance without considering all possible masks. Such measures can be obtained by approximating the change in training loss due to pruning, commonly through a Taylor expansion [Molchanov et al., 2017].

3.2.2 Taylor Expansion Analysis of Pruning Loss

The impact of pruning on the loss can be characterised by expanding the loss around the trained weights Θ^*. Let $\delta\Theta = M \odot \Theta^* - \Theta^*$ be the perturbation induced by masking:

$$\Delta\mathcal{L} \approx \underbrace{\nabla\mathcal{L}(\Theta^*)^\top \delta\Theta}_{\text{1st order}} + \frac{1}{2} \underbrace{\delta\Theta^\top H(\Theta^*)\, \delta\Theta}_{\text{2nd order}} + \mathcal{O}(\|\delta\Theta\|^3), \tag{3.3}$$

where $H(\Theta^*) \in \mathbb{R}^{P\times P}$ is the Hessian of the loss at Θ^*. The expansion separates the pruning effect into first-order sensitivity and second-order curvature (Eq. (3.3)). The first-order term measures how sensitive the loss is to the direction of weight removal: a weight with a large gradient, when removed, shifts the loss proportionally. What the second-order term adds is curvature. It captures how steeply the gradient itself changes as weights are zeroed out, which determines whether removing a small weight from a sharp loss valley is actually safe. At a stationary point of training where $\nabla\mathcal{L}(\Theta^*) \approx 0$, the first-order term vanishes and curvature becomes the decisive quantity.

Depending on which terms are retained and how H is approximated, this expansion gives rise to the full family of saliency criteria used in practice, from magnitude scoring (zeroth order, which ignores both terms and uses weight magnitude as a proxy) through gradient-based methods (first order) to blockwise curvature methods, each of which is developed in Section 3.4 and applied to specific methods in Chapters 5 and 4.

For second-order criteria, the same Taylor expansion can be used while keeping the curvature term. This gives a more accurate estimate of the loss change, but it also requires approximating the Hessian, which is expensive at modern scales. The original OBD criterion is a simple diagonal approximation, while later methods apply the same idea blockwise to recover tractability.

3.2.3 Optimal Brain Damage and Optimal Brain Surgeon

OBD [LeCun et al., 1990] is the simplest instantiation of the second-order term of the Taylor expansion (Eq. (3.3)): it assumes the Hessian is diagonal and applies no compensating correction to the remaining weights after the removal of a single weight, so the saliency of each weight reduces to

$$S_i^{\text{OBD}} = \frac{1}{2} H_{ii}\, \theta_i^2. \tag{3.4}$$

Under the OBD approximation, saliency becomes a Hessian-weighted square of the parameter (Eq. (3.4)). This makes ranking cheap, but it ignores interactions between weights. OBS [Hassibi & Stork, 1993] drops the diagonal assumption entirely: after each removal it solves for the optimal correction of the remaining weights induced by removing a weight, achieving strictly lower loss increase than OBD. The price is the full inverse Hessian, which is intractable at modern scales. This is where the connection to modern post-training compression becomes direct: later methods recover tractability by applying OBS-style updates blockwise, one layer at

a time, with the resulting per-layer formula derived in Section 3.4.3 [Frantar & Alistarh, 2023; Frantar et al., 2023].

3.2.4 Density, Sparsity, and Effective Topology

After a saliency criterion assigns scores to weights, the removal budget can be written through density and sparsity. Given N_{act} active (nonzero) parameters out of a total of P, the *density* and *sparsity* of the pruned network are

$$\rho = \frac{N_{\text{act}}}{P}, \qquad s = 1 - \rho. \tag{3.5}$$

Equation (3.5) gives the fraction of active parameters and the topology gives the placement of the zeros. At the same sparsity $s = 0.90$, the zeros may spread across layers, cluster in a few of them or form an irregular patterns that needs specialised sparse-format support [Hoefler et al., 2021]. For that reason, sparsity alone is insufficient to describe a compressed mode and must be paired with its *structure* of the removal pattern.

3.3 Pruning Structures

The method in which weights are removed will govern whether the resulting model can be accelerated in actual hardware, as well as how much leeway the pruning criterion has to choose among units to prune [Cheng et al., 2024a; Hoefler et al., 2021]. There are three structural regimes which subsume the fundamental choices available to pruning, all of which make their choices at different places on the trade-off curve between pruning flexibility and deployment efficiency:

3.3.1 Unstructured Pruning

Unstructured pruning removes individual scalar weights without any constraint on their positions in the weight matrix [Han et al., 2016]. Formally, given an importance score $S(\theta_i)$ for each parameter θ_i, the binary mask $M \in \{0,1\}^P$ retains the top-κ fraction of weights by score:

$$M_i = \mathbf{1}[S(\theta_i) \geq \tau_\kappa]\,, \tag{3.6}$$

where τ_κ is the $(1-\kappa)$ quantile of the score distribution. This connects the density constraint to a concrete mask rule: keep the weights above the score threshold and zero out the rest (Eqs. (3.1) and (3.6)).

This regime allows the most freedom since each weight gets its ownscore which usually means this gives the best accuracy for any level of sparsity[Frankle & Carbin, 2019]. This has cost at deployment as the final zero pattern is irregular and cannot be skipped efficiently without sparse storage and custom hardware that will store and only use the non-zero values and only perform the arithmetic for the non-zero weights (with special kernels for the skipping of zero arithmetic). In practice latencies improvements are not significant until a very high sparsity percentage (around 90%) and only in conjunction with sparse execution capabilities[Gale et al., 2020].

3.3.2 Semi-Structured Pruning ($N:M$ Sparsity)

$N:M$ ***sparsity*** is a constrained form of unstructured pruning that imposes a regular local pattern [Mishra et al., 2021]. Instead of allowing arbitrary weights to be removed anywhere in the matrix, the weight vector is partitioned into consecutive non-overlapping groups $\mathcal{G}$ of M elements. Within each group, exactly N elements are retained and the remaining $M - N$ are zeroed:

$$\|m_{\mathcal{G}}\|_0 = N \qquad \forall\, \mathcal{G},\ |\mathcal{G}| = M, \tag{3.7}$$

Here $m_{\mathcal{G}} \in \{0,1\}^M$ is the mask restricted to group $\mathcal{G}$ and $\|\cdot\|_0$ counts its nonzero entries. The constraint forces each local group to keep the same number of weights, which makes the sparsity pattern more predictable than unconstrained unstructured pruning.

A widely used instance of semi-structured sparsity is the 2:4 pattern, where only two weights are kept in every group of four weights [Cai, 2024; Y. Hu et al., 2024]. This gives the hardware a predictable sparse layout based on small weight groups. NVIDIA Ampere Sparse Tensor Cores support this fine-grained sparsity pattern and can exploit it for faster matrix multiplication when the required 2:4 constraint is satisfied [Mishra et al., 2021; NVIDIA Corporation, 2020].

After arbitrary weight removal and local group-wise constraints, a question that may come to mind is what happens if we try to remove coarser structures, such as complete architectural components. This is the topic of the next point.

3.3.3 Structured Pruning

Structured pruning removes entire vectors or submodules, producing a strictly smaller dense model with no irregular sparsity [H. Li et al., 2017]. Because the result is a standard dense matrix, no sparse storage format or specialised kernel is needed; the compressed model runs on any hardware that supports ordinary matrix multiplication.

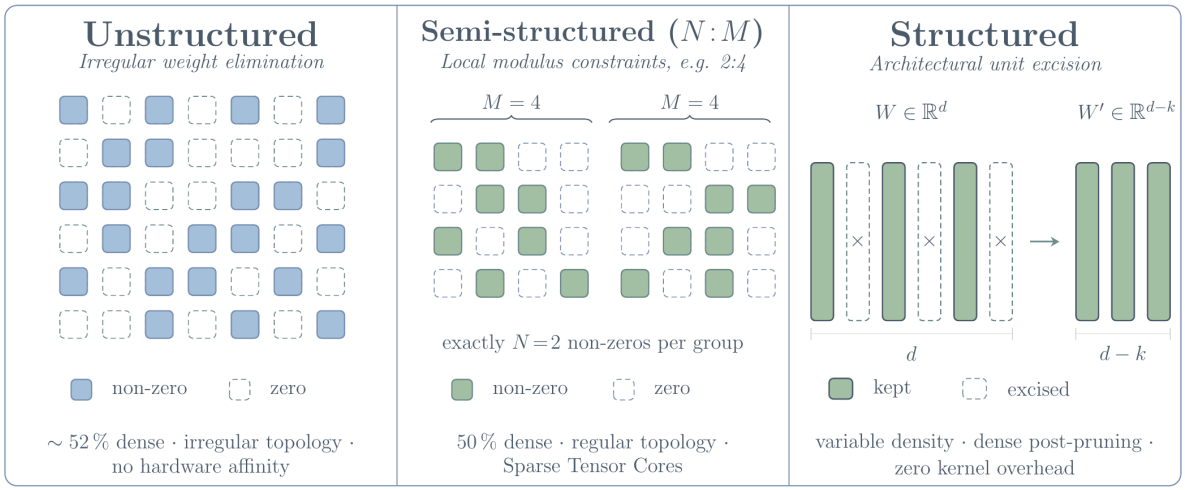
The removable unit depends on the architecture. In a convolutional network it is typically a filter or a channel; in a fully connected network it may be a neuron or a hidden dimension. The specific Transformer instantiation is covered later in Section 3.6.

Formally, removing rows and columns from a weight matrix $W \in \mathbb{R}^{m\times n}$ using a row mask $r \in \{0,1\}^m$ and a column mask $c \in \{0,1\}^n$ yields the submatrix

$$W' = W[\mathcal{R}, \mathcal{C}] \in \mathbb{R}^{m'\times n'}, \tag{3.8}$$

where $\mathcal{R} = \{i : r_i = 1\}$ and $\mathcal{C} = \{j : c_j = 1\}$ are the index sets of retained rows and columns, and $m' = |\mathcal{R}|$, $n' = |\mathcal{C}|$ are the reduced dimensions. The point of this construction is that structured pruning produces a smaller dense matrix, not a dense matrix with scattered zeros (Eq. (3.8)).

Figure 3.1 illustrates how the three regimes differ in their removal patterns. In the unstructured case, zeroed weights are scattered arbitrarily across the matrix, giving maximum freedom but producing no contiguous blocks that hardware can skip. Semi-structured sparsity resolves this by imposing a fixed retention ratio within each local group, aligning the pattern with dedicated hardware engines. Structured pruning goes further: removing entire rows or columns leaves a fully dense submatrix with no irregular zeros and no sparse-format overhead.



Property	Unstructured	$N:M$	Structured
Sparsity ratio	Any (up to 99%+)	Fixed (e.g. 50%)	Variable
Accuracy retention	Highest	High	Lowest
HW acceleration	Custom sparse kernels	Sparse Tensor Cores	Standard dense ops
Memory format	CSR / bitmap	2:1 compressed	Dense tensor

Figure 3.1: Comparison of unstructured, semi-structured, and structured sparsity patterns in a weight matrix.

3.4 Pruning Criteria

While the sparsity structure determines the shape and hardware footprint of the compressed model, it says nothing about *which* specific weights or units should be removed. That decision is governed by the pruning criterion: given a fixed removal budget, the criterion ranks the parameters or structures by their estimated importance to the network's output, so that the least important ones are removed first.

A ***pruning criterion*** assigns an importance score $S(\theta_i)$ (or $S(g)$ for a structured group g) that serves as a proxy for the loss change $\Delta\mathcal{L}$ (Eq. (3.3)) induced by removing that element. The ideal criterion ranks units by *true irreplaceability*: the unit with the lowest true loss impact should be removed first.

3.4.1 Magnitude-Based Criteria

The most common zeroth-order criterion evaluates a weight by its absolute value [Han et al., 2015]:

$$S_{\text{mag}}(w_i) = |w_i|. \tag{3.9}$$

For structured groups (e.g. a convolutional filter $W_g \in \mathbb{R}^{c_{\text{out}} \times c_{\text{in}} \times k \times k}$), the same idea is applied to the whole group. Common choices are the ℓ_1, ℓ_2, and ℓ_∞ norms:

$$S_g^{(1)} = \|W_g\|_1 = \sum_i |w_{g,i}|, \tag{3.10}$$

$$S_g^{(2)} = \|W_g\|_2 = \sqrt{\sum_i w_{g,i}^2}, \tag{3.11}$$

$$S_g^{(\infty)} = \|W_g\|_\infty = \max_i |w_{g,i}|. \tag{3.12}$$

These group scores extend the scalar magnitude rule to structured units: first score a whole tensor, then remove the groups with the smallest values. The ℓ_1 score measures total absolute magnitude, the ℓ_2 score measures Euclidean energy, and the ℓ_∞ score keeps only the largest entry in the group (Eqs. (3.9), (3.10), (3.11), and (3.12)).

The ℓ_2 norm is the most common because it is differentiable and rotation-invariant. The ℓ_∞ norm is sensitive to outliers and is less often used alone.

Interpretation : At a stationary point with a well-conditioned Hessian, $S_{\text{2nd}} = \frac{1}{2} H_{ii} w_i^2 \propto w_i^2$ if H_{ii} is approximately constant across parameters. Under this assumption, ranking by $|w_i|$ is equivalent to ranking by the second-order saliency [LeCun et al., 1990]. When Hessian diagonal entries vary greatly across layers (which they do in practice), magnitude alone may rank units from poorly-conditioned layers too aggressively.

3.4.2 Gradient-Based Criteria

First-order saliency estimates include the effect of the local gradient [Molchanov et al., 2017]:

$$S_{\text{grad}}(w_i) = \left| \frac{\partial \mathcal{L}}{\partial w_i} \cdot w_i \right|. \tag{3.13}$$

This is obtained from the first-order Taylor term of Eq. (3.3) with $\delta w_i = -w_i$ (complete removal), which turns the local gradient into a gradient-weight saliency score (Eq. (3.13)). For structured pruning, the same criterion can be extended to a removable group by summing over all constituent parameters:

$$S_{\text{grad}}(g) = \left| \sum_{i \in g} \frac{\partial \mathcal{L}}{\partial w_i} \cdot w_i \right|. \tag{3.14}$$

This produces one importance value for the whole group, not one value per scalar weight (Eq. (3.14)) [Molchanov et al., 2017].

3.4.3 Second-Order Criteria

For second-order criteria, the Taylor expansion (Eq. (3.3)) is kept to the curvature term, which gives a more accurate estimate of the loss change but requires approximating the Hessian. The original OBD criterion (Section 3.2.3) is a simple diagonal approximation, while later methods

apply the same idea blockwise to recover tractability. There are two main approaches to second-order criteria, both introduced in Section 3.2.3: the diagonal approximation of OBD and the blockwise inverse Hessian of OBS.

Diagonal approximation (OBD style).

$$S_{\text{2nd}}(w_i) = \frac{1}{2} H_{ii} w_i^2. \tag{3.15}$$

This score reproduces the OBD saliency of Eq. (3.4). The diagonal curvature term used in this score can be estimated from the Gauss-Newton approximation to the Hessian, which requires computing per-sample gradients:

$$H \approx G^\top G, \qquad G_{ij} = \frac{\partial \ell_j}{\partial w_i}, \tag{3.16}$$

where ℓ_j is the loss on sample j. The diagonal is then $H_{ii} \approx \sum_j G_{ij}^2$, giving the curvature factor used by the second-order score (Eq. (3.15)).

Blockwise inverse Hessian. The full OBS correction (Section 3.2.3) is too expensive to apply to all parameters at once, so modern methods apply the same idea locally inside large linear layers [Frantar & Alistarh, 2023; Hassibi & Stork, 1993]. In a layer with weight matrix $W \in \mathbb{R}^{m \times n}$, the calibration activations are collected in $X \in \mathbb{R}^{n \times N}$, with one column per calibration sample. These activations give the empirical input Hessian $\hat{H} = XX^\top/N$, which describes how input directions interact inside the layer. The layer is then processed column by column: when one weight is removed, the remaining weights in the current block are adjusted so that the layer output changes as little as possible.

$$W'_{:,j} = W_{:,j} - \frac{w_{:,j}}{[\hat{H}^{-1}]_{jj}} w_{:,j}^\top \hat{H}^{-1}_{j+1:,j}, \tag{3.17}$$

This correction, written in Eq. (3.17), carries the pruning error into the surviving columns instead of leaving it concentrated at the removed weight. After each removal, the remaining inverse Hessian is updated efficiently, so the next pruning decision is made with the current block state. The total cost is $O(n^3 + mn^2)$ per layer.

3.4.4 Activation-Aware Criteria

Some criteria scale saliency by the activation statistics at the weight's input. Concretely, the weight magnitude is multiplied by the ℓ_2 norm of the input feature it processes [Sun et al., 2024]:

$$S_{\text{act}}(W_{i,j}) = |W_{i,j}| \cdot \|x_j\|_2 , \tag{3.18}$$

where $\|x_j\|_2$ is the ℓ_2 norm of the j-th input feature computed over a calibration set of N samples calculated as shown in Eq. (3.19):

$$\|x_j\|_2 = \sqrt{\sum_{s=1}^{N} x_{s,j}^2} . \tag{3.19}$$

This equals $\sqrt{N}$ times the root-mean-square (RMS) of that feature's activations. RMS measures how much energy a feature carries on average across inputs. A feature with consistently large activations has high RMS. One that is rarely triggered has low RMS and contributes little to the output regardless of its associated weight's magnitude. Equation (3.18) makes the criterion sensitive to both: a small weight on a high-RMS feature scores higher than a larger weight on a near-silent one.

Activation magnitudes are distributed unevenly across input features [Sun et al., 2024; Wei et al., 2022]. Magnitude-only scoring ignores this. It assigns identical importance to a weight on a high-energy feature and to an equally large weight on a near-zero feature, systematically underestimating the contribution of the first group. Scaling by $\|x_j\|_2$ corrects that bias directly.

3.4.5 Learnable and Adaptive Criteria

A fourth family introduces *trainable mask scores* $v \in \mathbb{R}^P$ that are optimised jointly with weights [Louizos et al., 2018; Mallya & Lazebnik, 2018]:

$$M_i = \sigma\left(\frac{v_i - \bar{v}}{\epsilon}\right), \quad \text{or} \quad M_i = \mathbf{1}[v_i \geq \tau], \tag{3.20}$$

The mask can be kept soft during training or converted into a hard keep/remove decision (Eq. (3.20)). The discrete threshold is approximated during backpropagation by a straight-through estimator [Bengio et al., 2013], which passes the gradient through the threshold as if it were the identity. The learning objective then adds a sparsity penalty so that accurate but dense masks are discouraged:

$$\mathcal{L}_{\text{sparse}} = \mathcal{L}(\Theta, M(v)) + \lambda \|v\|_1 . \tag{3.21}$$

This objective balances task performance against mask sparsity (Eq. (3.21)).

These approaches are valuable especially in the setting where the training distribution is accessible and the budget for mask optimisation is not the constraint. At moderate sparsities in fine-tuning, they outperform static criteria [Louizos et al., 2018; Sanh et al., 2020].

3.4.6 Criterion Comparison

Table 3.2 places the criteria developed in this section side-by-side. Reading down the table means to scan one trade-off: more primitive criteria (top) just see the weights. More sophisticated criteria (bottom) requires gradients, curvature, activations, or a training signal.

Table 3.2: Summary comparison of pruning criteria.

Criterion	Information used	Data needed	Grad?	H?	Typical accuracy
Magnitude	Weight values only	None	No	No	Baseline
Gradient (Taylor)	Gradient × weight	Train/calib	Yes	No	+1–3% over mag
OBD (§3.2.3)	Diag. curvature	Train/calib	Yes	Diag	+1–5% over mag
Blockwise OBS (§3.4.3)	Full curvature per layer	Calibration	No	Block	Best (post-train)
Activation-weighted	Magnitude × activation	Calibration	No	No	Strong post-train
Learned / trajectory mask	Score trajectory	Fine-tune	Yes	No	Best for fine-tune

3.5 Pruning Strategies

The criteria defined in Section 3.4 state *what* to remove. A related question is *when* this removal occurs and when to let the model recover in between removals. A pruning schedule describes the pattern of mask updates and recoveries during the course of training or preparing for inference, which is what this section covers.

3.5.1 One-Shot Pruning

In ***one-shot pruning***, importance scores are computed once from the fully trained model and all removal decisions are made in a single pass [Han et al., 2016]. Algorithm 1 shows the procedure.

Algorithm 1: One-Shot Pruning

Input: Trained model $f(\cdot; \Theta)$, target density κ
Output: Pruned model $f(\cdot; M \odot \Theta)$
Compute $S(\theta_i)$ for all $i \in [P]$
Set threshold $\tau \leftarrow (1 - \kappa)$ quantile of $\{S(\theta_i)\}$
$M_i \leftarrow \mathbf{1}[S(\theta_i) \geq \tau]$
Optional: fine-tune Θ with M fixed
return $M \odot \Theta$

One-shot pruning does not need any additional training rounds and it can be applied directly to a pretrained network, so it is easy to run. However, this simplicity does not come free. All the removal decisions are derived from one pass through the loss surface using a small calibration set without letting the model to recover after each mask update. The accuracy for a given sparsity is not as good as iterative techniques [Han et al., 2016; Zhu & Gupta, 2018].

3.5.2 Iterative Pruning and Recovery

Iterative pruning directly addresses the limitation of one-shot pruning by interleaving mask updates with weight recovery steps [Zhu & Gupta, 2018]. Instead of removing all targeted weights at once, the model is pruned gradually over R rounds, with a fine-tuning phase after each round to let the remaining weights compensate for what was removed. Algorithm 2 shows a common formulation using a cubic sparsity schedule.

Algorithm 2: Iterative Pruning

Input: Trained model Θ, target density κ, rounds R
Output: Pruned model $M \odot \Theta$

$\kappa_0 \leftarrow 1.0$ `// start fully dense`
for $r = 1$ **to** R **do**
  $\kappa_r \leftarrow \textsc{Schedule}(\kappa,\ r,\ R)$ `// interpolate density toward target`
  Compute scores $S(\theta_i)$ on current Θ `// re-evaluate importance`
  Update mask M to density κ_r `// remove lowest-scoring units`
  Fine-tune Θ with M fixed for T_r steps `// recover accuracy`
end
return $M \odot \Theta$

The SCHEDULE function can take many forms such as linear, cubic, cosine, or step, and the choice is often borrowed directly from learning rate scheduling practice, since both problems share the same goal of controlling how aggressively a quantity changes over the course of training [Zhu & Gupta, 2018]. Figure 3.2 illustrates several common variants.

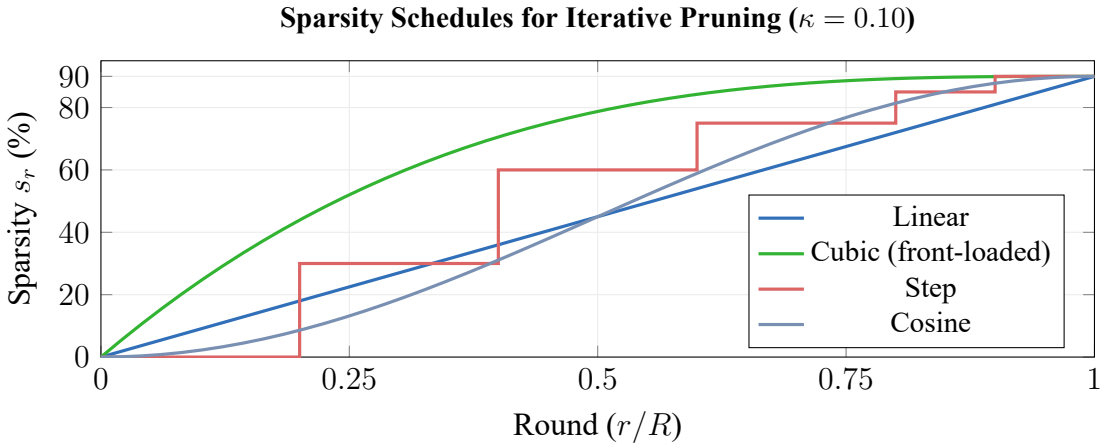


Figure 3.2: Variants of sparsity schedules

Linear growth increases each round by the same amount, meaning that the final removal rate of training will be the same as the initial rate, and there is no settling at the final dense rate. Cubic growth, in contrast, puts the majority of the pruning effort at the beginning of training; by the time r/R is 0.5, already about 87% of the remaining budget of high sparsity is spent, and remaining rounds are held at near-final density, only increasing pruning rate negligibly between them.

Step schedules perform discrete steps at fixed intervals and the mask is held between the steps.

Cosine growth, after initial slow rate of increase, grows quickly through the middle, before again beginning to slow as sparsity is nearing the target. [Zhu & Gupta, 2018].

3.5.3 Pruning During Training

A third family involves sparsity directly within the optimisation loop by adding a regularisation term (often an L1 norm [Liu et al., 2017]) to the training loss. The L1 penalty pushes weights towards zero continually during the whole training process which consequently makes sparsity evolve during the entire training and hard-pruning only happens after that point [Guo et al., 2016; Han et al., 2016]. The only condition of this method is having access to the whole training distribution, hence it can be mostly applied at training time from scratch or during supervised fine-tuning.

3.5.4 Post-Training Pruning

Post-training pruning applies pruning to a frozen pretrained model using only a small calibration set, without any gradient updates to the original training loss [Frantar & Alistarh, 2023]. This makes it the practical default for large pretrained networks, where full retraining over the original training distribution is prohibitively expensive.

The procedure can be separated in 4 steps. Firstly, the smallest calibration set (several samples) are propagated through the frozen model, the statistics of layer-wise input activation are recorded. Secondly, these statistics are combined with the weight values, in order to calculate a saliency score $S(\theta_i)$ for every single weight or group using one of the pruning criteria of Section 3.4. Thirdly, the weights with the lowest scores are removed, to produce the binary mask M^*. Finally, an optional step which can be included: the values of the remaining weights in every layer are correct, in order to compensate for the damage resulted by the removals; this concept has been inherited from OBS (Section 3.4.3, [Hassibi & Stork, 1993]) and is used blockwise in the state of the art post-training method [Frantar & Alistarh, 2023]. Figure 3.3 summarizes this process.

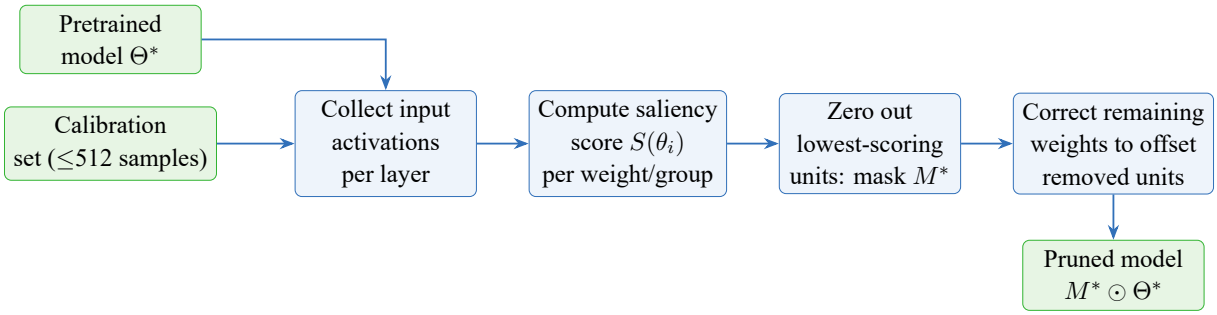


Figure 3.3: Post-training pruning pipeline.

So far, we have described pruning by means of three components: what structure should be eliminated, according to which evaluation measure, and which application strategy should be adopted. Because our specific interest is Transformers, these components now have to be identified within the architecture being compressed. In the next section, we therefore describe an instantiation of the pruning approach within Transformer models.

3.6 Pruning Transformers

The general framework described so far identifies three decisions: which units to remove (Section 3.3), by what criterion (Section 3.4), and according to which schedule (Section 3.5). Applying this to a Transformer means mapping each decision to the specific computational structures of the architecture. The three main removable units are attention heads, FFN channels, and entire Transformer layers. Each carries a different cost and requires a different adaptation of the pruning formulas.

3.6.1 Attention Head Pruning

The multi-head attention module computes attention in H parallel subspaces called heads. Given an input sequence $X \in \mathbb{R}^{n\times d}$, where n is the sequence length and d is the model dimension, each head h projects X into query, key, and value spaces of dimension $d_h = d/H$ using learned matrices $W_{Q_h}, W_{K_h}, W_{V_h} \in \mathbb{R}^{d\times d_h}$, computes scaled dot-product attention, and feeds the result through the shared output projection $W_O \in \mathbb{R}^{d\times d}$.

A head mask $m \in \{0,1\}^H$ selects which of the H heads contribute to the output:

$$\text{MHA}_m(X) = \text{Concat}(m_1\,\text{head}_1(X), \ldots, m_H\,\text{head}_H(X))\,W_O. \tag{3.22}$$

Setting $m_h = 0$ zeros out W_{Q_h}, W_{K_h}, W_{V_h}, and the corresponding block of columns in W_O (Eq. (3.22)). Since each head uses query, key, value, and output projections, the number of parameters removed per head is:

$$\Delta P_{\text{head}} = 4\,d\,d_h = \frac{4d^2}{H}. \tag{3.23}$$

Early work on head importance found that a large share of heads can be removed with no statistically significant drop in performance [Michel et al., 2019; Voita et al., 2019]. Heads that carry out identifiable functions, such as attending to adjacent tokens or tracking syntactic dependencies, are harder to remove, while heads whose attention patterns closely resemble those of other heads in the same layer are the most dispensable [Voita et al., 2019]. Importance varies considerably across layers and no single head is universally critical [Michel et al., 2019].

3.6.2 FFN Channel Pruning

The feed-forward network in each Transformer layer expands the model dimension by a factor r before projecting back. For an input $X \in \mathbb{R}^{n\times d}$ and expansion ratio r, the standard FFN applies:

$$\text{FFN}(X) = \phi(XW_1 + b_1)\,W_2 + b_2, \tag{3.24}$$

where $W_1 \in \mathbb{R}^{d\times rd}$, $W_2 \in \mathbb{R}^{rd\times d}$, and ϕ is a pointwise activation function. This is the standard two-projection FFN block used before channel pruning (Eq. (3.24)). A channel mask $g \in \{0,1\}^{rd}$ selects which of the rd intermediate neurons to retain:

$$\text{FFN}_g(X) = \phi(XW_1 + b_1)\,\text{diag}(g)\,W_2 + b_2. \tag{3.25}$$

Writing $\mathcal{K} = \{k : g_k = 1\}$ for the retained channel indices, the masked FFN can be rewritten as a dense FFN of smaller width:

$$\text{FFN}_g(X) = \phi(XW_{1,\mathcal{K}} + b_{1,\mathcal{K}})\, W_{2,\mathcal{K},:} + b_2. \tag{3.26}$$

Removing one channel eliminates one row of W_1 and one column of W_2, which is exactly what the masked and compact forms express (Eqs. (3.25) and (3.26)). Each removed channel therefore saves $2d$ parameters. For $rd - |\mathcal{K}|$ removed channels the total saving is $(2d)(rd - |\mathcal{K}|)$.

3.6.3 Layer Pruning

Each Transformer block is an additive function: $F_l(x) = x + F_l^{(1)}(x)$, where $F_l^{(1)}(x)$ is the composition of the attention sub-layer and the FFN sub-layer. When $F_l^{(1)}(x)$ is small relative to x, the residual connection carries the bulk of the signal and the block contributes little to the representation. Removing it leaves downstream activations largely unchanged.

A natural way to measure this is to compare the magnitude of $F_l(x)$ to the magnitude of the input x over a calibration set. Using the Frobenius norm $\|\cdot\|_F$ (the square root of the sum of all squared entries of a matrix) as the measure of size, the relative block contribution is:

$$S_l^{\text{block}} = \frac{\|F_l(X)\|_F}{\|X\|_F}, \tag{3.27}$$

where the norms are computed over a batch of calibration inputs X. A block with a small relative contribution is close to an identity map, so it becomes a natural candidate for removal (Eq. (3.27)) [Kurtić et al., 2023; Michel et al., 2019].

Layer pruning is the crudest pruning mechanism due to its impact on the whole model: when a layer is removed, it not only removes all the parameters in one attention module but also those in one FFN. These constitute significantly more parameters compared to a removed head or a set of pruned channels. While efficient in decreasing the latency of inference (since latency grows linearly with the number of layers), fine-tuning is usually required if more than a couple of layers are removed [Kurtić et al., 2023].

3.7 Conclusion

This chapter formally posed pruning as a constrained optimization problem and discussed the three regimes of structural sparsity, comparing the saliency criteria for magnitude pruning down to second-order blockwise. Section 3.6 discusses the instantiation of these concepts for the Transformer. The pruning strategy is critical for determining when the mask should be updated and the degree of recovery between mask update iterations; whether the pruning occurs one-shot, iteratively, during or post-training all result in a different accuracy-cost curve.

Together these components provide the vocabulary the method-level review in Chapters 4 and 5 builds on.

Chapter 4

Related work on pruning: Unstructured & Heterogeneous Methods

4.1 Introduction

In chapter 3 we defined the formal problem of pruning, described the most common saliency measures and the structural regimes of sparsity. In the context of this theory, the pruning literature can be viewed as an ongoing debate among successive more elaborate answers to the problem of ”how to discard computation without compromising the behavior of the reference model under specific deployment constraints”.

Since there is such a large and diverse body of pruning work, we break down our survey of the literature on methods into two chapters, according to the induced sparsity topology. Chapter 4 details Unstructured and Heterogeneous methods, whereas Chapter 5 details Structured pruning methods. Before presenting the methods themselves, we introduce an organized taxonomy to help place these approaches.

4.2 Taxonomy of pruning methods

The main sparsity topologies were introduced formally in Section 3.3 (see Figure 3.1). From a deployment standpoint, they differ in whether pruning yields compressed storage alone or native dense-shape acceleration after model surgery.

The taxonomy adopted here follows four coupled axes. The *removable unit* determines whether the method acts on scalar weights or executable structures such as attention heads, FFN channels, and layers [Kurtic et al., 2022; Michel et al., 2019]. The *optimization signal* ranges from magnitude and activation statistics to differentiable mask learning, first-order Taylor criteria, second-order curvature surrogates, or search-based objectives [Frantar & Alistarh, 2023; Kwon et al., 2022; Ma et al., 2023; Sun et al., 2024; Xia et al., 2022]. The *recovery stage* distinguishes direct post-training pruning from methods requiring LoRA recovery, gradual fine-tuning, or distillation. The *deployment objective* determines whether the reported benefit is compressed storage, dense-shape acceleration, or measured speedup [Kurtić et al., 2023; Sieberling

et al., 2025; Tang et al., 2025].

Each family corresponds to a different deployment objective. Unstructured methods such as Wanda and SparseGPT primarily optimize weight-level saliency [Frantar & Alistarh, 2023; Sun et al., 2024]. Structured methods span differentiable mask-learning (CoFi), dependency-aware gradient methods (LLM-Pruner), Fisher-based post-training frameworks, and runtime-aware second-order methods (ZipLM) [Kurtić et al., 2023; Kwon et al., 2022; Ma et al., 2023; Xia et al., 2022]. Search-based methods such as EvoPress and DarwinLM treat pruning as configuration optimization under explicit budget or hardware constraints [Sieberling et al., 2025; Tang et al., 2025]. Figure 4.1 and Table 4.1 summarize this organization.

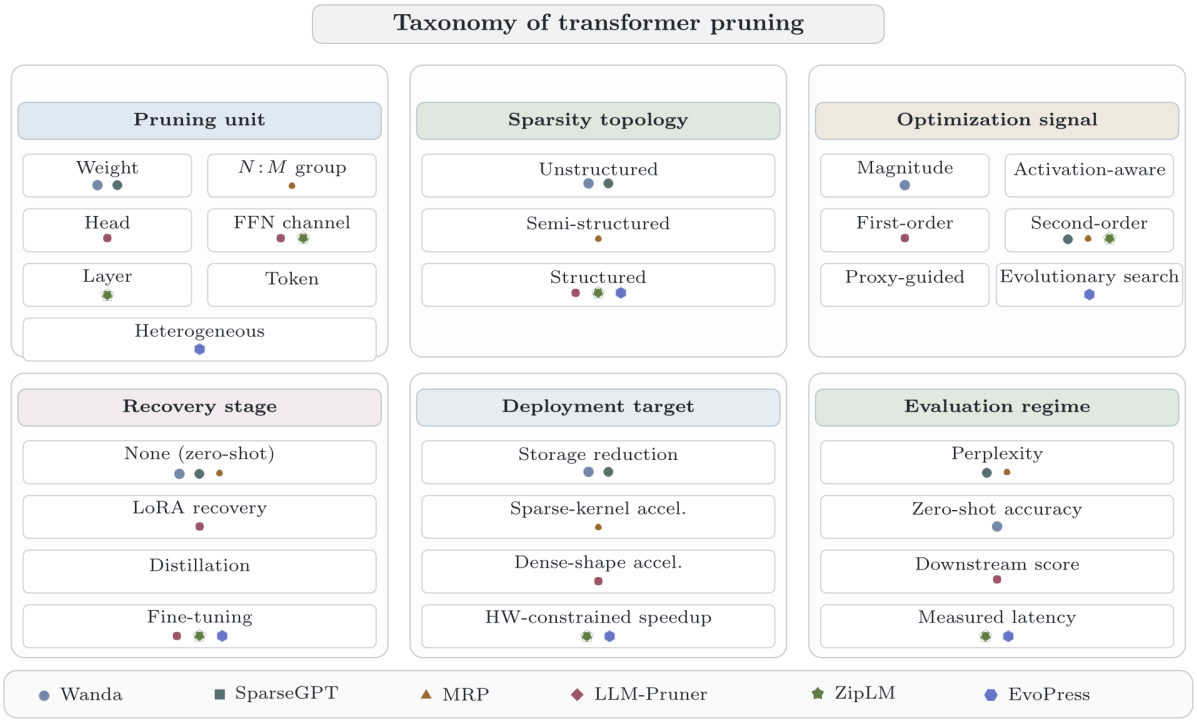


Figure 4.1: Classification of pruning methods by the unit of removal, sparsity topology, optimization signal, recovery stage, and primary deployment claim

Table 4.1: Summary of pruning methods by unit, topology, signal, recovery, and deployment claim

Method	Unit	Topology	Signal	Recovery	Primary deployment claim
Wanda	Weight	Unstructured	Activation-aware saliency	None	Storage-oriented sparsity with strong zero-shot perplexity retention
SparseGPT	Weight	Unstructured	Single-weight second-order compensation	None	High-sparsity pruning with local reconstruction
CoFi	Layers / heads / FFN / hidden dimensions	Structured	Learned hard-concrete masks with L_0 control and distillation	Full task-specific pruning and fine-tuning	Dense-shape acceleration through jointly pruned Transformer structure
LLM-Pruner	Heads / channels / blocks	Structured	First-order Taylor scoring with dependency groups	LoRA recovery	Structural compression with dense-shape execution
Fast post-training pruning	Heads / FFN filters	Structured	Fisher-based constrained mask optimization	None beyond mask tuning	FLOPs- or latency-constrained structured pruning without retraining
ZipLM	Heads / FFN dimensions	Structured	Second-order structured saliency with runtime-constrained search	Gradual fine-tuning and distillation	Measured speedup under device-specific constraints
EvoPress / DarwinLM	Layer-wise compression levels	Heterogeneous structured	Proxy-guided or evolutionary search	Search-time evaluation or proxy training	Global configuration search under budget and hardware constraints

Figure 4.1 separates the pruning literature by the unit being removed and by the metric used to decide removal. Table 4.1 then makes this map concrete by assigning each reviewed method to a pruning unit, topology, scoring mechanism, recovery regime, and deployment claim.

Guided by this taxonomy, the remainder of this chapter reviews methods that prioritize maximum theoretical flexibility and search space exploration: Unstructured and Heterogeneous Search-based methods (Wanda, SparseGPT, EvoPress, and DarwinLM). Structured methods are reviewed in Chapter 5.

4.3 Saliency-Based Unstructured Methods

Remember that in Section 3.4 the Taylor expansion of the loss change for a weight θ_i that is pruned is given by equation (3.3). It is made up of a first-order term proportional to the gradient, and second-order terms relating to the curvature. This category removes both and approximates using the magnitude of the weight or an activation statistic surrogate of the true change, and requires no gradient and no calibration except the forward pass, rendering them the least expensive in our taxonomy.

Magnitude and Activation-Aware Baselines

Weight-Magnitude Pruning. Magnitude pruning uses absolute weight values as the saliency proxy:

$$\mathcal{S}_{i,j}^{\text{mag}} = |W_{i,j}|. \tag{4.1}$$

Its core assumption is that small weights contribute less to the output and can be removed with limited degradation. The method requires no gradient-based recovery and remains the simplest baseline against which more sophisticated algorithms are evaluated.

Wanda (Weights $\times$ Activations). Wanda [Sun et al., 2024] refines this baseline by incorporating activation information from a small calibration set. The saliency score is

$$\mathcal{S}_{i,j}^{\text{wanda}} = |W_{i,j}| \cdot \|x_j\|_2, \tag{4.2}$$

where x_j is the j-th input activation channel and $\|x_j\|_2$ is its RMS norm estimated over N calibration samples. The key insight is that importance is jointly encoded in weight magnitude and the energy of the coupled input channel.

Both magnitude pruning and Wanda are one-shot post-training methods in the regime of Section 3.5.4: scores are computed once and no recovery training is applied afterward. Wanda processes each layer independently: it collects input activations on a small calibration set (typically 128 samples), computes per-weight saliency scores as the product of weight magnitude and activation norm, then prunes the lowest-scoring weights for each output neuron until the target sparsity p is reached. No weight update or recovery step is applied. Magnitude pruning is a pure linear scan $O(|W|)$ with no calibration; Wanda adds only an activation-collection pass, keeping the overall cost near-linear.

These saliency methods are useful because they are cheap and easy to apply, but their decisions remain local. Once pruning moves from individual weights to heads, FFN channels, or layers, the method must account for structural dependencies between parameters. This motivates gradient-based structured pruning methods, where the score is computed over groups that can be removed consistently.

4.4 Second-Order Unstructured Methods & Semi-Structured Methods

While magnitude and Wanda are purely zeroth-order, second-order methods such as SparseGPT [Frantar & Alistarh, 2023] and MRP [Kwon et al., 2022] attempt to capture curvature information to improve pruning decisions. SparseGPT estimates the Hessian of the loss with respect to weights using a block-wise approximation and applies a closed-form compensation to the remaining weights after pruning. MRP extends this idea to N:M sparsity patterns, where groups of weights are pruned together, and updates the compensation iteratively as more groups are removed. These methods are more expensive than local scoring but can achieve higher accuracy at high sparsity levels by accounting for the interactions between pruned and retained weights.

However, they still operate at the weight level and do not consider the structural dependencies that arise when pruning entire heads, channels, or layers. This motivates the next family of methods that optimize for structured pruning with gradient-based criteria and recovery.

4.4.1 Optimal Brain Compression Formulation

For a layer with weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and calibration activations $\mathbf{X} \in \mathbb{R}^{d_{\text{in}} \times n}$, the layer-wise sparse reconstruction problem is

$$\min_{\mathbf{W}^{\text{sparse}}} \|\mathbf{W}\mathbf{X} - \mathbf{W}^{\text{sparse}}\mathbf{X}\|_F^2 \quad \text{s.t.} \quad \|\mathbf{W}^{\text{sparse}}\|_0 \leq (1-p)\, d_{\text{out}} d_{\text{in}}, \tag{4.3}$$

where $\|\cdot\|_F$ is the Frobenius norm, $\|\cdot\|_0$ counts the number of nonzero entries, and $p \in (0,1)$ is the target sparsity ratio. Using the empirical Hessian proxy $\mathbf{H} = \mathbf{X}\mathbf{X}^T/n$, this becomes a curvature-aware reconstruction problem whose solution depends on $\mathbf{H}^{-1}$ to redistribute pruning error across remaining weights. The methods that follow differ in how they instantiate this objective: SparseGPT removes one scalar weight at a time, MRP removes an entire N:M group jointly, Fisher-based pruning applies curvature scoring to structured masks, and ZipLM extends the same principle to executable Transformer structures under runtime constraints.

4.4.2 Single-Weight Iterative Removal (SparseGPT)

SparseGPT [Frantar & Alistarh, 2023] greedily prunes one weight at a time while applying compensation updates to remaining weights in the current block. This can be read as a blockwise version of the OBS correction introduced in Section 3.2.3 and formalised for large linear layers in Section 3.4.3. Its local objective at iteration t is

$$\min_{\delta w} \frac{1}{2} \delta w^T H \delta w \quad \text{s.t.} \quad (w + \delta w) \cdot e_t = 0, \tag{4.4}$$

where e_t is the one-hot selector for the pruned weight.

SparseGPT is a post-training one-shot method in the regime of Section 3.5.4: pruning and compensation are applied in a single forward pass over the layers without any subsequent gradient-based recovery. It processes the weight matrix in column blocks of size B. For each block, it first constructs the regularized inverse Hessian $(XX^T + \lambda I)^{-1}$. Within each block, it iteratively identifies the weight with minimum local second-order loss, sets it to zero, then updates the remaining weights in that block using the corresponding column of $\mathbf{H}^{-1}$. After processing a block, lazy batch updates are applied to adjacent blocks. The method scales as $O(d_{\text{in}}^2 d_{\text{out}} + d_{\text{in}}^3/B)$ per layer, with a calibration budget of approximately 128 samples.

It is more instructive to understand SparseGPT as a weight-level, Hessian-compensated method for pruning: each decision to prune weights is made locally, and weights are updated to make the layer output's deviation as minimal as possible [Frantar & Alistarh, 2023]. It is through this same lens that SparseGPT's modification for semi-structured N:M sparsity can be understood, where the compensation method is kept and mask selection is constrained to local groups. If the block size is set to $B_s = M$, and from each group of M weights the N weights

with the smallest OBS reconstruction score are selected for removal, the compensation becomes more localized to a block instead of applying it for the whole layer. Within that block, however, weights are still selected and compensated one at a time, so the compensation at each step does not account for the interactions among the other weights being simultaneously removed from the same group, which is the gap MRP directly addresses.

4.4.3 Joint Optimization for Multi-Weight Removal (MRP)

The MRP framework [Huang et al., 2025] generalizes second-order pruning to N:M groups. For a vectorized weight block w and binary pruning mask M, where $M_i = 1$ indicates that weight i is pruned, the objective minimizes the second-order loss increase for the full group:

$$\min_{\delta w} \frac{1}{2}\delta w^T H \delta w + g^T \delta w \quad \text{s.t.} \quad (w + \delta w) \odot M = 0, \tag{4.5}$$

where g is the gradient of the loss with respect to w, and $\odot$ denotes elementwise multiplication. Solving the group jointly yields a compensation update for the retained weights:

$$\delta w^\star = -H^{-1}_{\bar{M}\bar{M}}(g_{\bar{M}} + H_{\bar{M}M} w_M), \tag{4.6}$$

where M denotes pruned indices, $\bar{M}$ retained indices, and w_M the weight values at the pruned coordinates. This expression makes the role of the block Hessian explicit: in N:M pruning, the removed weights are not independent decisions, so compensation must be computed for the group.

MRP is a post-training one-shot method in the regime of Section 3.5.4: no gradient updates or recovery training are applied after pruning. It operates layer by layer. For each layer, the block Hessian is estimated from calibration activations. The weight tensor is partitioned into non-overlapping groups of M consecutive weights. For each group, the joint second-order problem is solved: the mask selecting the $M-N$ weights with the highest reconstruction cost is identified, and the compensation update $\delta w^\star$ is applied to the retained N weights before the next group is processed. Because the group is treated as a group, the method correctly accounts for interactions among simultaneously pruned coordinates within the block.

Figure 4.2 illustrates this difference. SparseGPT compensates after removing one scalar weight, whereas MRP compensates after removing a constrained local group.

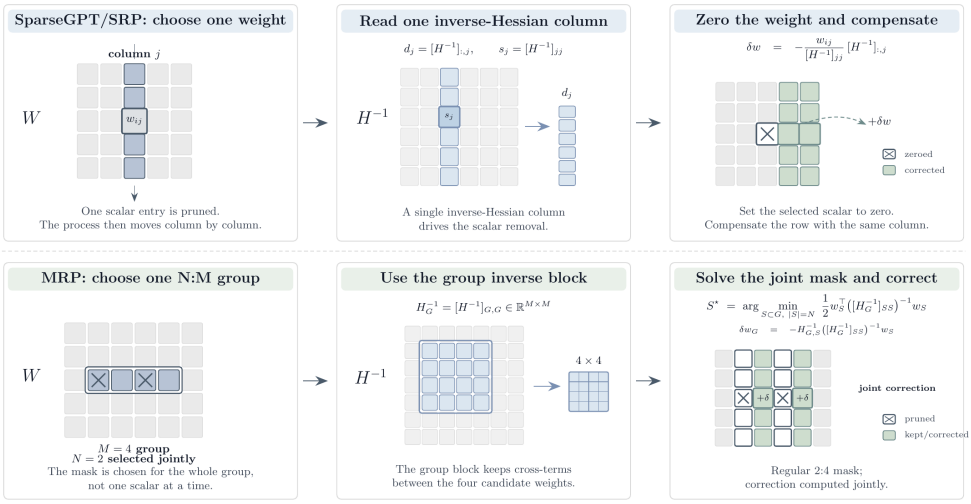


Figure 4.2: Second-order compensation strategies for unstructured and semi-structured pruning.

4.5 Search-Based Heterogeneous Methods

Search-based methods are formulated over entire sets of sparsity configurations, rather than applying a single closed-form ranking rule per unit as in the cases above. This is necessary when a pruning decision quality is contingent on the interaction between several deletions across multiple layers. Search-based methods are less efficient than local scoring methods but have a far wider search space and discover higher quality candidates at high levels of compression. The two exemplary methods described above are EvoPress, an evolutionary search over discrete layer-wise pruning configurations, and DarwinLM which expands this over structured pruning using a train-aware selection scheme.

EvoPress. EvoPress [Sieberling et al., 2025] searches over discrete layer-wise compression profiles instead of assigning a single saliency score per layer. A candidate is a configuration $\mathbf{C} = \{c_1, \ldots, c_L\}$ where $c_\ell \in \mathcal{S}_\ell$ is the compression level for layer ℓ. Budget is preserved by level-switch mutation: one layer becomes more compressed only if another becomes less compressed, keeping the global average sparsity S_{glob} fixed. Mutations are deliberately local, with most offspring differing from the parent by only one or two switches.

Selection proceeds in multiple rounds with increasing token budgets and decreasing survivor counts:

$$\mathcal{P}^{(r+1)} = \text{Top}_{n_{r+1}} \left(\left\{ \mathcal{F}_{T_r}(C) \mid C \in \mathcal{P}^{(r)} \right\} \right), \qquad T_1 < T_2 < \cdots, \; n_1 > n_2 > \cdots, \tag{4.7}$$

where the fitness signal is KL divergence to the reference model. This progressive evaluation scheme is what makes EvoPress practically efficient: early rounds use few tokens to discard poor candidates quickly, reserving the full calibration budget for finalists. The method converges in approximately 5–6 GPU hours for a standard Llama-2-7B sparsity search.

EvoPress is a one-shot, post-training search method (Sections 3.5.1 and 3.5.4): candidate profiles are scored only by their divergence to the reference model, and the weights themselves

are never updated or fine-tuned during the search.

Figure 4.3 summarizes this search loop. The important point is that the algorithm evaluates complete compression profiles, selects promising candidates with increasing evaluation budgets, and mutates the retained profile while preserving the global compression constraint.

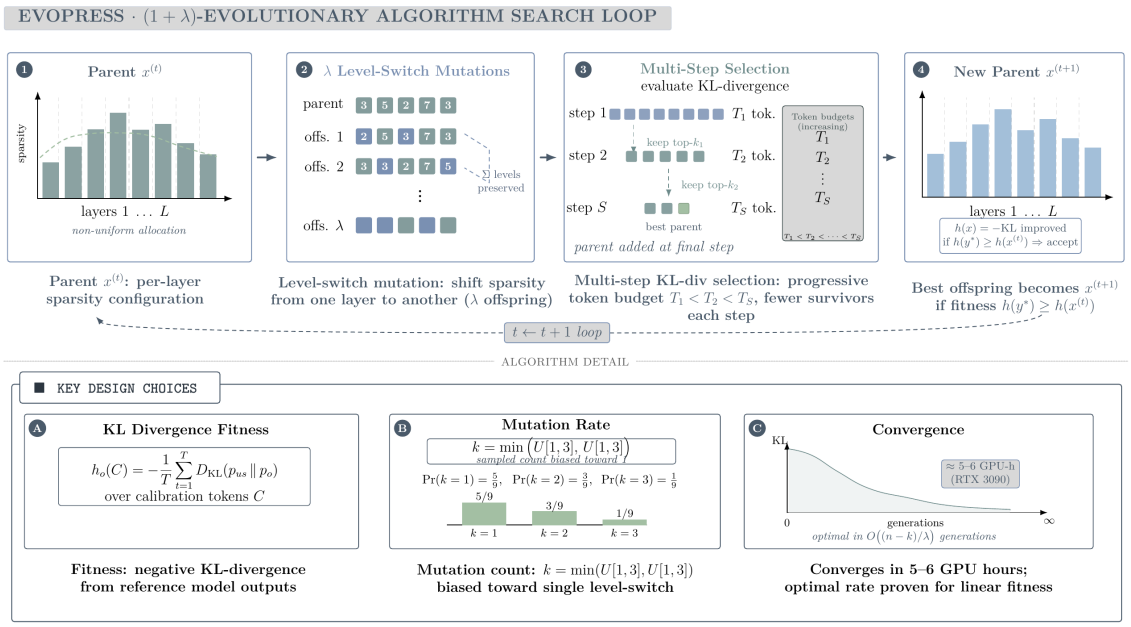


Figure 4.3: EvoPress $(1 + \lambda)$ evolutionary search loop for dynamic LLM compression [Sieberling et al., 2025]. The method applies budget-preserving level-switch mutation and progressively evaluates candidates with larger calibration budgets.

DarwinLM. DarwinLM [Tang et al., 2025] extends evolutionary search to *structured* pruning by unifying OBS-style second-order pruning with a training-aware offspring selection strategy. It differs from EvoPress on three axes: structured granularity (heads and FFN channels), front-loaded Hessian computation stored in a sparsity-level database (SLD), and fitness measured after a short lightweight finetune.

For each module, DarwinLM solves the structured OBS problem at N_l equally-spaced sparsity levels and stores the pruned-and-compensated weights. The optimal structured mask and compensation are:

$$M^* = \arg\min_{M} \sum_{i=1}^{d_{\text{row}}} \mathbf{W}_{i,M} \left(\hat{H}^{-1}_{MM}\right)^{-1} \mathbf{W}^{\top}_{i,M}, \tag{4.8}$$

$$\delta^* = -\mathbf{W}_{:,M} \left(\hat{H}^{-1}_{MM}\right)^{-1} \hat{H}^{-1}_{M,:}, \tag{4.9}$$

where d_{row} is the output dimension of the module weight matrix, M indexes the pruned columns, and $\hat{H}$ is the empirical Hessian computed from calibration activations. During search, candidate models are assembled by looking up stored entries, avoiding any Hessian recomputation.

Each candidate receives a short lightweight finetune before evaluation:

$$\mathcal{F}(C_k) = D_{\mathrm{KL}}\big(\hat{M} \parallel \text{finetune}(C_k,\, T_f)\big), \tag{4.10}$$

where $\hat{M}$ is the output distribution of the dense reference model and T_f is the fine-tuning token budget. Selection uses three progressive rounds ($T_f \in \{10\text{K}, 50\text{K}, 200\text{K}\}$ tokens) with decreasing survivor counts. The SLD must be computed once per model and sparsity range, making the upfront cost proportional to $L \cdot N_l$ structured OBS operations. DarwinLM compresses Llama-2-7B to 2.7B parameters with higher downstream accuracy than ShearedLlama using $5\times$ fewer post-training tokens (10B vs. 50B).

DarwinLM sits between a post-training regime (Section 3.5.4) and an iterative-recovery regime (Section 3.5.2): the SLD is built offline without end-to-end training, but each search candidate receives a lightweight fine-tune before fitness evaluation. In the offline phase, the SLD is built: for each module and each of N_l sparsity levels, the structured OBS problem is solved and the pruned-and-compensated weights are stored, so that any candidate configuration can be assembled by table lookup. In the search phase, candidate models are stitched together by selecting one SLD entry per module, then evaluated through that lightweight finetune. Mutation follows the same budget-preserving logic as EvoPress, swapping one module's sparsity level up only if another's is swapped down. Selection proceeds over three rounds with token budgets of 10K, 50K, and 200K and shrinking survivor counts, so weaker candidates are eliminated cheaply before the full evaluation budget is spent. The upfront SLD cost is proportional to $L \cdot N_l$ structured OBS operations and is paid once per model.

Figure 4.4 shows the corresponding DarwinLM workflow. Compared with EvoPress, the key visual distinction is the front-loaded sparsity-level database: candidate models are assembled from stored structured-pruning choices, then filtered through training-aware selection.

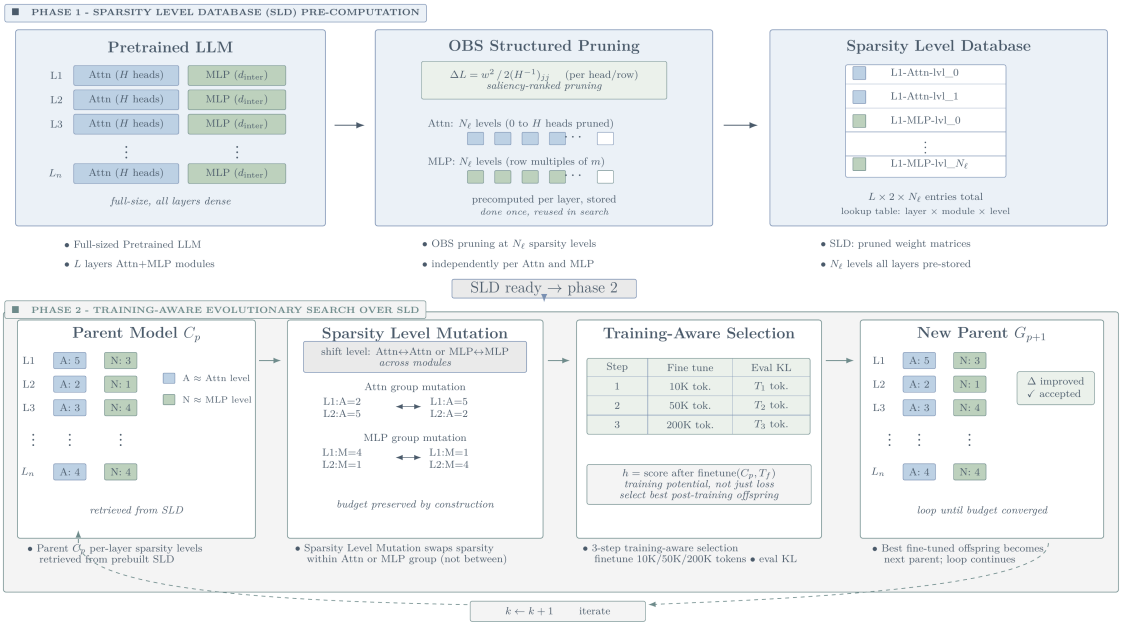


Figure 4.4: DarwinLM search loop [Tang et al., 2025]. The method first builds a sparsity-level database with OBS-style structured pruning, then performs training-aware evolutionary selection over stitched candidate models.

Table 4.2 compares the practical preparation cost of the reviewed families.

Table 4.2: Preparation cost and evaluation requirements of representative pruning methods. The table separates low-cost post-training scoring, training-dependent structured pruning, curvature-based reconstruction, and population-based search.

Method	**Dominant preprocessing cost**	**Additional requirement**
Magnitude	Linear scan $O(\|W\|)$	None
Wanda	Activation collection and linear scoring	Small calibration set ($\approx$128 samples)
CoFi	Multi-epoch joint mask and weight optimization	Task-specific teacher; $\approx$33 h for MNLI
LLM-Pruner	Forward/backward scoring over dependency groups	LoRA recovery ($\approx$2.5 h on RTX 4090)
SparseGPT	Block-wise Hessian estimation and reconstruction	Calibration Hessian
MRP	Group-wise mask search and compensation	Block curvature updates for N:M groups
Fast PTPF	Diagonal or block Fisher estimation with mask tuning	Calibration set; optional latency lookup table
ZipLM	Layer-wise Hessian updates and runtime-table search	Calibration set plus hardware benchmarking pass
EvoPress / DarwinLM	Population search over compression profiles	1–6 GPU-hours; SLD pre-computation for DarwinLM

4.6 Conclusion

In this chapter, we surveyed the methods that leverage theoretical flexibility, exploration of search space; we called them Unstructured, Heterogeneous Search-based pruning methods. In unstructured pruning, such as Wanda, SparseGPT, only weigh-level saliency is exploited. In search-based pruning methods like EvoPress, DarwinLM, the pruning problem is cast as finding the optimal configuration subject to explicit budgets or hardware constraints.

Such methods are more interesting at high-sparsity, when interaction among various removed elements through different layers may significantly affect accuracy. Although more

expensive than local scoring methods, they explore a broader search space and are able to find a better performing solution at high-sparsity albeit using more computation for pruning. The next chapter is going to introduce structured methods, those optimize on dense-shape acceleration by simultaneously removing executable structures like attention heads, FFN channels, and layers.

Chapter 5

Related work on pruning: Structured Methods

5.1 Introduction

In Chapter 4 we considered pruning of individual weights or at different, irregular sparse levels. However, such pruning can only achieve actual speed-ups in reality if there is specific sparse-execution hardware which supports it. Chapter concentrates on the other side of our classification taxonomy. Structured Pruning.

Instead of removing a few weight matrices throughout a given weight matrix by scattering zeros, structured pruning removes larger chunks of code, such as a single attention head, a set of feed forward channels or an entire Transformer layer. By taking such building blocks out of the model, the pruned model has exactly the same form as the original one: still a dense neural network, just with fewer units. Such networks can be sped up on standard hardware with an out-of-the-box speedup through optimized dense matrix multiplications.

However, completely eliminating units results in heavy destruction of forward pass; therefore the criterion of removal would be far more complex than unstructured weights' magnitude. According to the structure described in the taxonomy above, this chapter will surveys four state-of-the-art methods that handle structured pruning without destructive removal-LLM-Pruner, CoFi, Fast PTPF, ZipLM by dependency graph, trainable masks and 2 nd order optimization methods.

5.2 Gradient-Based Methods

The first order term of the Taylor expansion (Equation (3.3)) retains a value that is a product of the gradient and the weight value. The criterion thus becomes sensitive to the direction of loss change as well as the weight magnitude. The methods in this section utilize precisely this first order signal, however instead of single values it utilizes grouped structure, for example heads, channels and blocks.

Structured pruning replaces individual-weight saliency with group-level importance, because the natural units in a Transformer, namely attention heads, FFN channels, and entire layers, are coupled through residual pathways and shared projection matrices. Removing one head, for example, requires deleting consistent slices from W_Q, W_K, W_V, and W_O together; removing an FFN channel removes coordinated rows and columns from the two projection matrices. Any pruning criterion applied at this level must therefore identify groups that can be removed as a unit without breaking the forward pass [Kurtic et al., 2022; Michel et al., 2019].

LLM-Pruner. LLM-Pruner [Ma et al., 2023] addresses the dependency problem explicitly by constructing a graph over the neurons of the network. For neurons N_i and N_j with input set $\text{In}(N)$ and output set $\text{Out}(N)$, the graph encodes two rules:

$$N_j \in \text{Out}(N_i) \wedge \deg^-(N_j) = 1 \implies N_j \text{ depends on } N_i, \tag{5.1}$$

$$N_i \in \text{In}(N_j) \wedge \deg^+(N_i) = 1 \implies N_i \text{ depends on } N_j. \tag{5.2}$$

Here $\deg^-(N_j)$ is the in-degree of N_j (number of neurons feeding into it) and $\deg^+(N_i)$ is the out-degree of N_i. The transitive closure of these rules groups all coupled parameters into a single dependency group $\mathcal{G} = \{W_i\}_{i=1}^{M}$, which is the minimal unit that can be consistently removed.

Once groups are defined, each is scored using a first-order Taylor approximation of the loss increase over a small calibration set $\mathcal{D}$. The importance of a structure tensor W_i is estimated as

$$I_{W_i} \approx \left| \frac{\partial \mathcal{L}^\top(\mathcal{D})}{\partial W_i} W_i - \frac{1}{2} W_i^\top H W_i \right|, \tag{5.3}$$

where H is the local Hessian. The first-order gradient is retained alongside the Hessian term because the calibration set is only a proxy for the pretraining distribution; discarding it would introduce bias. An element-wise variant replaces H with a Fisher-diagonal approximation. Individual tensor scores within a group are aggregated by sum, product, max, or last-only operators, and groups with the lowest aggregate score are removed first.

LLM-Pruner sits between a post-training and a recovery regime: scoring is post-training and requires no gradient propagation through the full model, but the pruned network is then passed through a LoRA recovery stage in line with the iterative recovery approach of Section 3.5.2. LoRA adapters [E. J. Hu et al., 2022] are inserted into all linear modules and fine-tuned for two epochs over roughly 50K samples. The low-rank factors are then merged back into the weight matrices, leaving no adapter overhead at inference time. The full scoring stage requires only 10 short calibration sequences, making the preprocessing cost modest relative to the recovery phase.

CoFi. CoFi [Xia et al., 2022] is the main structured-pruning example of the learnable-mask criteria introduced in Section 3.4.5. It places binary masks on attention blocks, heads, FFN blocks, intermediate channels, and the shared hidden dimension, then optimises these masks

jointly with the task loss:

$$\mathrm{MHA}(X) = z_{\mathrm{MHA}} \sum_{i=1}^{N_h} z_{\mathrm{head}}^{(i)} \,\mathrm{Att}\Big(W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}, W_O^{(i)}, X\Big)\,, \tag{5.4}$$

$$\mathrm{FFN}(X) = z_{\mathrm{FFN}}\,\mathrm{gelu}(XW_U)\,\mathrm{diag}(z_{\mathrm{int}})\,W_D, \tag{5.5}$$

where N_h is the number of attention heads, $W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}, W_O^{(i)}$ are the query, key, value, and output projection matrices of head i, and W_U, W_D are the up- and down-projection matrices of the FFN. Each mask variable z is relaxed to a continuous value during training using the hard-concrete distribution for L_0 regularisation, which allows gradients to flow through the mask. The training objective combines task loss with a distillation term and a Lagrangian sparsity constraint:

$$\mathcal{L}_{\mathrm{prune}} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{distil}} + \lambda_1(\hat{s} - s_0) + \lambda_2(\hat{s} - s_0)^2, \tag{5.6}$$

where s_0 is the sparsity target and $\hat{s}$ is the expected sparsity under the current mask distribution. Because the masks at different granularities interact, CoFi can discover structured subnetworks that a simple per-group ranking would miss.

CoFi is a training-time iterative method in the sense of Section 3.5.2 where mask decisions and weight updates are interleaved throughout training. A warm-up phase first stabilizes the model weights before mask learning begins, with the sparsity constraint applied softly. The main phase then jointly optimises masks and weights under the Lagrangian objective, progressively tightening the sparsity target toward s_0 over training. Once the target is reached, each continuous mask variable is thresholded against zero to produce a binary decision, and a final fine-tuning pass is run on the surviving subnetwork. Because masks at all granularities are trained simultaneously.

Gradient-based structured pruning improves the consistency of the removed units, but the score still depends on a local approximation of how removal affects the loss. When pruning becomes more aggressive, interactions among remaining and removed weights become more important. This leads to second-order methods, which use curvature information or local reconstruction objectives to compensate for the error introduced by pruning.

5.3 Second-Order Structured Methods

Gradient-based structured pruning improves the consistency of the removed units, but the score still depends on a local first-order approximation. When pruning becomes more aggressive, interactions among remaining and removed weights become more important. To compensate for the error introduced by structural removal, the methods in this section apply the second-order principles (such as Fisher information and inverse Hessians) introduced in the previous chapter, but adapted specifically for executable Transformer blocks.

5.3.1 Fisher-Based Post-Training Structured Pruning

The post-training framework of Kwon et al. [Kwon et al., 2022] applies second-order structured pruning without end-to-end post-pruning weight optimization. Binary masks over atten-

tion heads and FFN filters are found by solving

$$\arg\min_{\mathbf{m}} \mathcal{L}(\mathbf{m}) \quad \text{s.t.} \quad \text{Cost}(\mathbf{m}) \leq C, \tag{5.7}$$

where Cost is either a FLOPs or measured latency budget. Expanding around the dense mask and applying a diagonal Fisher approximation reduces the search to selecting units with the smallest diagonal Fisher scores subject to the cost constraint. For latency-constrained pruning, a piecewise-linear lookup table replaces the analytical cost model to incorporate real device behavior:

$$\text{LAT}(\mathbf{m}_\ell) = \begin{cases} 0, & \|\mathbf{m}_\ell\|_0 = 0, \\ c, & 0 < \|\mathbf{m}_\ell\|_0 \leq T, \\ a(\|\mathbf{m}_\ell\|_0 - T) + c, & \|\mathbf{m}_\ell\|_0 > T, \end{cases} \tag{5.8}$$

Where c represents the base latency for any non-empty layer settings, T denotes the structural threshold below which the latency remains relatively constant, and a is the slope describing how measured latency increases beyond T. The lookup table is accessed when searching for the mask to get an estimate for the latency of each pruned MHA or FFN layer on the target device, enabling the selection of a mask that fulfills a measured latency budget.

The method proceeds in four steps: (1) diagonal Fisher scores are computed for head and FFN-filter masks on a small calibration set (2K training examples); (2) a FLOPs- or latency-constrained mask search is solved under the diagonal Fisher objective; (3) a block-diagonal Fisher refinement is applied layer by layer to recover interactions ignored by the diagonal approximation; (4) surviving mask values are tuned by layer-wise linear least-squares reconstruction. Because only mask values are updated, the method avoids end-to-end weight retraining. It is therefore a one-shot, post-training method (Sections 3.5.1 and 3.5.4): the mask is selected in a single constrained pass and only mask values are tuned, with no iterative prune–recover cycle. The paper reports pruning times of 39 seconds on GLUE and 135 seconds on SQuAD, compared to 33 hours for CoFi on MNLI.

5.3.2 Inference-Aware Structured Second-Order Pruning (ZipLM)

ZipLM[Kurtić et al., 2023] extends the OBS-style reconstruction objective from individual weights to executable Transformer structures. The pruned objects are attention heads, FFN intermediate channels, and, at the coarser level, entire residual modules such as the attention block or the FFN block. For a candidate structure S with column mask $\mathbf{M}_S$, the removal score is

$$\arg\min_{S} \sum_{i=1}^{d_{\text{row}}} \mathbf{W}_{i,\mathbf{M}_S} \left(\left(\mathbf{H}^{-1} \right)_{\mathbf{M}_S,\mathbf{M}_S} \right)^{-1} \mathbf{W}_{i,\mathbf{M}_S}^T, \tag{5.9}$$

where d_{row} is the output dimension of the weight matrix and the sum aggregates the reconstruction cost across all output rows. After removing a structure, the exact compensation and Hessian update are applied through block Gaussian elimination, so that all subsequent saliency scores remain consistent with the already-pruned state.

ZipLM adds a device-level objective on top of the local second-order scoring. It benchmarks attention and FFN configurations on the target hardware and stores measured runtimes in a

lookup table. The global search then minimizes layer-wise structural sensitivity subject to a real speed constraint $\sum_{\ell} \tau_{\ell,s_\ell} \leq T_{\text{target}}$.

ZipLM follows an iterative pruning-and-recovery regime in the sense of Section 3.5.2: pruning steps are interleaved with fine-tuning rounds. It operates in two phases (see Figure 5.1). In the global phase, attention-head and FFN configurations are benchmarked on the target environment and a layer-wise sparsity profile satisfying the speedup requirement is selected. In the local phase, each selected layer is compressed by iteratively scoring candidate structures with the structured saliency metric, applying the optimal weight compensation, and updating the inverse Hessian by block Gaussian elimination. Recovery follows a gradual prune-and-fine-tune schedule with combined task, logit, and token-level distillation losses, at a cost of 8–10 fine-tuning epochs per pruning step.

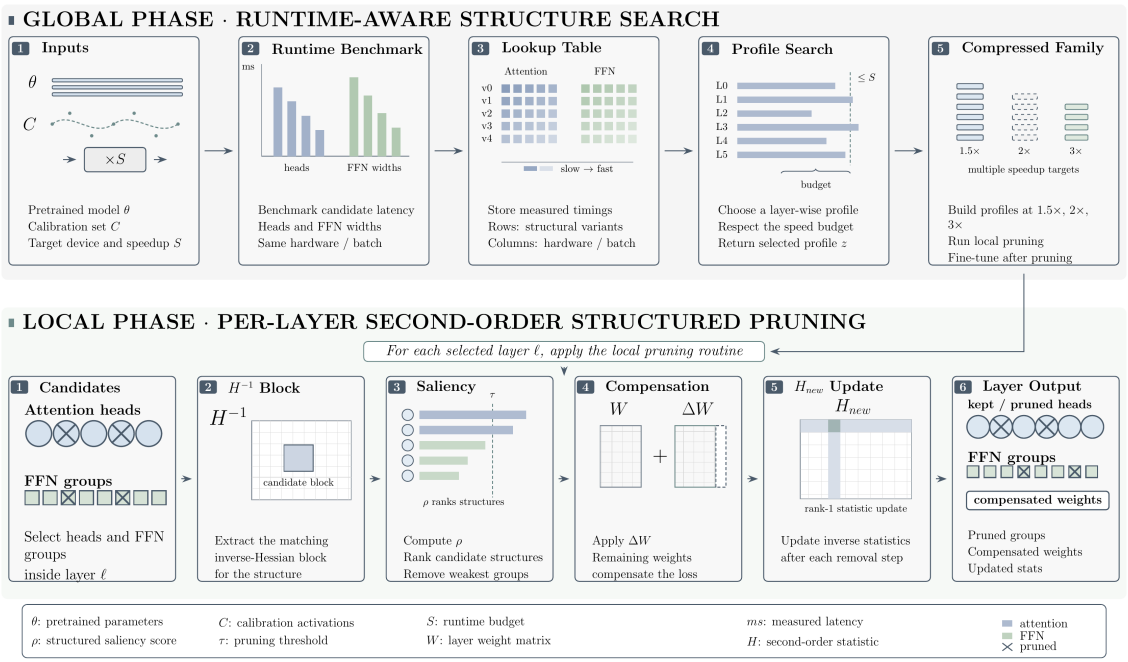


Figure 5.1: ZipLM pipeline. The global phase benchmarks attention-head and FFN configurations and searches for layer-wise sparsity profiles satisfying the required speedup. The local phase applies second-order structured pruning through inverse-Hessian scoring, compensation updates, and iterative Hessian refinement.

Second-order methods reduce local reconstruction error, but they do not fully resolve the allocation problem across the whole model. A method may know how to remove a head or channel with limited local damage and still not know how many heads or FFN channels should remain in each layer under a global budget. This is the point at which search-based methods become relevant.

5.4 Conclusion

In this chapter, we showed how structured pruning can be used to produce hardware-agnostic dense-shape acceleration. Using LLM-Pruner, CoFi, Fast PTPF and ZipLM as examples, we demonstrated how pruning objectives should evolve to accommodate highly coupled structures such as attention heads and feed-forward channels. While structural pruning can be highly destructive, these methods incorporate complex techniques to maintain performance from dependency modeling to co-design of masks and second-order Hessian approximations.

These individual results, however, cannot be taken at face value without understanding how they perform relative to one another. To properly grasp the state-of-the-art in Transformer compression, we need to compare these methods that involve the structures presented in this chapter with the unstructured and heterogeneous techniques developed in Chapter 4. The next chapter therefore introduces a rigorous cross-regime empirical comparison, characterizes the major shortcomings of today's evaluation protocols, and provides a formal description of the open problem of "allocation" that inspires this thesis's contribution.

Chapter 6

Synthesis, Empirical Comparison, and Limitations

6.1 Introduction

Chapter 4 introduced a survey on Transformer pruning with unstructured/heterogeneous techniques, and Chapter 5 described pruning with structured techniques. Due to their differences in terms of optimized units, recovered pipelines and optimization objective metrics, the results of unstructured pruning and structured pruning are hard to be compressed into one table, which implies that we cannot provide a "golden" result directly applicable. The discussion in this chapter is concentrated on what the real evidence supports: in which circumstances pruning itself is responsible for the compression, the result is more related with the recovering mechanism, or it is difficult to find out an appropriate global structure with local score mechanism.

6.2 Empirical Comparison

The empirical results are grouped by pruning regime. Unstructured and N:M methods mainly report perplexity under weight sparsity, structured methods report task accuracy or speedup after architectural removal, and search-based methods report complete pruning configurations under budget constraints.

6.2.1 Unstructured and Semi-Structured Results

Table 6.1 reports representative low, medium, and large model settings. The unstructured rows compare magnitude pruning, SparseGPT, and Wanda at 50% sparsity, and the 2:4 rows show the semi-structured setting that motivates group-aware compensation such as MRP.

Table 6.1: Representative unstructured and semi-structured pruning results on WikiText perplexity. PPL denotes perplexity; lower is better.

Model	Setting	Dense PPL	Magnitude PPL	SparseGPT PPL	Wanda PPL
LLaMA-7B	50% unstructured	5.68	17.29	7.22	7.26
LLaMA-30B	50% unstructured	4.77	7.54	5.31	5.24
LLaMA-65B	50% unstructured	3.56	5.90	4.57	4.57
OPT-13B	50% unstructured	10.13	1.2×10^4	11.17	–
OPT-175B	50% unstructured	8.35	4.3×10^4	8.21	–
OPT-13B	2:4 semi-structured	10.13	–	12.96	–
OPT-66B	2:4 semi-structured	9.34	–	10.09	–
OPT-175B	2:4 semi-structured	8.35	–	8.74	–

The results show that pure magnitude pruning can fail badly at 50% sparsity, especially on OPT models, whereas activation-aware and second-order criteria preserve perplexity much more effectively. The 2:4 rows also show that semi-structured pruning is not just unstructured pruning with a different mask shape: the local group constraint changes the compensation problem and motivates methods such as MRP.

6.2.2 Structured Pruning Results

Table 6.2 summarizes representative structured-pruning results across compression and speedup regimes.

Table 6.2: Representative structured Transformer pruning results across recovery regimes. GLUE denotes the General Language Understanding Evaluation benchmark, SQuADv1 denotes Stanford Question Answering Dataset version 1.1, acc. denotes accuracy, and F1 denotes question-answering F1 score.

Method	Model	Setting	Key result
LLM-Pruner [Ma et al., 2023]	LLaMA-7B	20% prune ratio	94.97% zero-shot accuracy retention after LoRA recovery
	LLaMA-7B	50% prune ratio	77.4% zero-shot accuracy retention after LoRA recovery
CoFi [Xia et al., 2022]	BERT Base	GLUE/MNLI, $\approx 12\times$ speedup	80.6 acc.; TinyBERT$_4$ reports 78.8 at similar speedup
	BERT Base	SQuADv1, $\approx 8.7\times$ speedup	82.6 F1, matching TinyBERT$_4$ at comparable speedup
Fast PTPF [Kwon et al., 2022]	BERT Base	GLUE and SQuADv1, FLOPs-constrained pruning	60–70% FLOPs with $\leq$ 1% accuracy/F1 drop
	BERT Base	GLUE and SQuADv1, latency-constrained pruning on V100	$1.34\times$–$1.47\times$ geometric-mean speedup depending on batch size
ZipLM [Kurtić et al., 2023]	BERT Base	GLUE/QNLI and MNLI, $2\times$ speedup	91.4 / 84.8 acc.; essentially dense-level performance
	BERT Base	GLUE/QNLI and MNLI, $10\times$ speedup	88.6 / 82.7 acc.; approximately 5M parameters
	BERT Large	SQuADv1, $10\times$ speedup	89.1 F1 with 20.9M parameters

These results are closer to deployment because the methods remove executable units, but they still depend heavily on the evaluation and recovery setup. CoFi uses task-specific training, LLM-Pruner depends on LoRA recovery, Fast PTPF is largely post-training, and ZipLM adds

device-specific runtime profiling. The numbers therefore show the achievable trade-offs within each protocol, not a clean ranking across methods.

6.2.3 Evolutionary and Search-Based Results

Table 6.3 reports representative search-based results for EvoPress and DarwinLM. EvoPress searches over depth-pruning and sparsity-allocation configurations, and DarwinLM uses structured evolutionary pruning with training-aware offspring selection.

Table 6.3: Representative search-based pruning results. Wiki2 and C4 are perplexity values where lower is better; task average is accuracy where higher is better.

Setting	Model	Budget	Method	Wiki2↓	C4↓	Task Avg.↑
Depth pruning	Mistral-7B	25%	EvoPress	**8.66**	**12.04**	–
	Mistral-7B	25%	ShortGPT	43.26	40.16	–
	Mistral-7B	25%	Shortened LLaMA	14.94	19.30	–
Depth pruning	Llama-2-7B	37.5%	EvoPress	**17.98**	**18.91**	–
	Llama-2-7B	37.5%	ShortGPT	70.94	63.51	–
	Llama-2-7B	37.5%	Shortened LLaMA	35.37	26.07	–
Unstructured allocation	Mistral-7B	70%	EvoPress	**14.42**	**16.46**	**53.8**
	Mistral-7B	70%	OWL	17.22	21.66	51.9
	Mistral-7B	70%	Uniform	23.08	30.03	49.9
Structured search	Llama-2-7B	2.7B model	DarwinLM	–	–	62.8 after 10B-token post-training
	Llama-2-7B	2.7B model	ShearedLLaMA	–	–	62.6 after 50B-token post-training

The search-based results make the allocation issue explicit. EvoPress improves over local or heuristic allocation when multiple removals interact, and DarwinLM shows that training-aware evolutionary selection can produce strong structured models with less post-training data than a fixed pruning baseline.

6.2.4 Cross-Regime Synthesis

Table 6.4 condenses these comparisons into the main implications for pruning-search design.

Table 6.4: Aggregated interpretation of the empirical pruning evidence by regime.

Regime	Representative setting	Selected evidence	Reading
Weight-level unstructured	LLaMA-7B, 50% sparsity	Dense PPL 5.68; magnitude 17.29; SparseGPT 7.22; Wanda 7.26.	Better saliency preserves perplexity, but the resulting pattern remains unstructured and does not guarantee dense-kernel speedup.
Semi-structured N:M	OPT-13B, 2:4 sparsity	Dense PPL 10.13; SparseGPT-style compensation gives 12.96 under the 2:4 constraint.	N:M sparsity needs group-aware compensation, which motivates MRP-style joint treatment of simultaneously removed weights.
Structured LLM pruning	LLM-Pruner on LLaMA-7B	20% prune ratio retains 98.6% zero-shot accuracy after LoRA recovery; 50% retains 83.6%.	Dependency-aware pruning keeps the model executable, with final quality strongly tied to recovery.
Structured encoder pruning	CoFi, Fast PTPF, and ZipLM on BERT	CoFi reaches 80.6 MNLI acc. near 12× speedup; Fast PTPF keeps 60–70% FLOPs with $\leq$1% drop; ZipLM reaches 88.6/82.7 QNLI/MNLI acc. at 10× speedup.	Structured pruning can produce executable acceleration, with protocols differing in training cost, hardware profiling, and recovery budget.
Search-based pruning	EvoPress and DarwinLM	EvoPress improves depth-pruning and sparsity-allocation baselines; DarwinLM reports 62.8 average accuracy after 10B-token post-training versus 62.6 for ShearedLLaMA after 50B tokens.	Configuration search addresses global allocation, and training-aware selection improves structured candidates beyond one-shot scoring.

6.3 Discussion

The distinction between the two questions has been separated between Chapters 4 and 5 and in the empirical comparison, Section 6.2. One question is: for a single unit that is potentially removable, what criterion should be used to rank the units? Here, different magnitude-based, gradient-based, and curvature-based criteria offer varying estimates. The second question is: how is the overall budget of pruning shared among the units in the model? Here, the approaches tend to diverge more substantially, and much of the outstanding debate in the literature is focused on the latter. It will be discussed here, then the empirical data.

6.3.1 Local Scoring Versus Global Allocation

The methods reviewed in this chapter differ in implementation, pruning unit, and recovery cost, but they can all be related to the same underlying objective:

$$\min_{\mathcal{M}\in\mathcal{F}} \Delta\mathcal{L}(\mathcal{M}), \tag{6.1}$$

where $\mathcal{M}$ denotes a feasible sparsity pattern or structured subnetwork and $\Delta\mathcal{L}$ denotes the loss increase caused by pruning. The difficulty is that $\mathcal{M}$ must satisfy a compression budget while preserving the behaviour of the model.

Another persistent issue with the pruning criteria discussed so far is their evaluation of weights and activations, which can be removed before the final compressed model is fixed. SparseGPT calculates approximate second-order local reconstruction costs of sets of weights [Frantar & Alistarh, 2023], while Wanda sorts parameters by a product of weight magnitude

and input activation norm [Sun et al., 2024]. These methods are successful in finding some locally removable weights but fail to determine the overall distribution of compression under a fixed budget across the layers [Cheng et al., 2024b; Hoefler et al., 2021]. Though one-shot local pruning works well under low compression ratios, high compression rates require more sophisticated iterative and global-allocation-based pruning methods [Janusz et al., 2025]. Several of these recent methods implicitly or explicitly address this issue: ZipLM attempts to find explicit local and global correlations between layers, and EvoPress directly searches over the layer-wise compression budgets that minimize the distribution divergence of the output [Kurtić et al., 2023; Sieberling et al., 2025].

6.3.2 Limitations of Local Ranking at High Compression

In the analyses above, we can observe one trend across these method families. It appears that local ranking criteria are successful when the compression ratio is relatively small to medium; when the targeted compression ratio becomes higher, the deficiencies of local ranking criteria tend to become apparent [Bai et al., 2024; Hoefler et al., 2021; Janusz et al., 2025; Sanh et al., 2020]. This pattern is expected from the definition of the pruning problem: local criteria judge removeable units while the ultimate compressed model has not been formed completely.

For Transformer models, structured pruning is naturally expressed as a layer-wise allocation problem over executable units. A pruned Transformer may retain different numbers of attention heads, FFN intermediate dimensions, hidden dimensions, or even complete MHA/FFN modules in different layers [Cheng et al., 2024b; Kurtić et al., 2023; Xia et al., 2022]. Thus, for a model with L layers, H attention heads per layer, and F FFN channels per layer, the retained structure can be represented by an allocation vector

$$\mathbf{s} = (h_1, f_1, h_2, f_2, \ldots, h_L, f_L),$$

subject to a global MAC budget Ω. Recent layer-wise pruning work therefore treats compression as the problem of assigning adaptive sparsity ratios across layers which is just a specific case of non uniform pruning [L. Li et al., 2024].

Local ranking simply assigns each prunable unit a saliency and prunes the lowest ones until the budget is exhausted. it implicitly models global ranking with the assumption that each pruning decision can be made in isolation. EvoPress recognizes this assumption in the domain of depth pruning; block-scoring algorithms use the assumption that error is linear such that each block is thought to contribute equally to the overall compression error [Sieberling et al., 2025]. ZipLM takes the reverse argument, stating that pruning has to capture both within layer local correlations and across layer global correlations [Kurtić et al., 2023].

The compositional structure of the Transformer violates the independence approximation. Each Transformer block comprises the residual-connected attention and feed-forward sublayers and self-attention layers take representations as input which are generated from previous layers [Vaswani et al., 2017]. With regard to pruning, this indicates that any alteration on one layer will bring new input distribution for the following layers. SparseLLM is the first one who precisely identifies the problem by modeling an LLM as a chain of modules, where outputs of a module

become the inputs for another, and points out that local layer error reduction only causes non-optimal performance for the global pruning at a high sparsity rate [Bai et al., 2024].

The same coupling is present for attention and the global budget constraint. Scaled dot-product attention normalizes the query-key score using a softmax, while multi-head attention merges multiple heads which act on the same input layer [Vaswani et al., 2017]. Experimentally, attention heads can have unequal importance, numerous can be dropped, beyond a threshold, attention collapses catastrophically, and different components of the attention function can have differing sensitivity [Michel et al., 2019]. In addition, if the MAC budget is fixed, having high capacity in one layer means we must sacrifice capacity elsewhere. Prior works have indicated the necessity of across-layer uneven sparsity under high compression [Gale et al., 2019; L. Li et al., 2024; Sanh et al., 2020].

6.4 Empirical Results

Having argued that allocation is more important than local scoring at high compression, the question is whether the results reflect this. On the whole, they do but only if we treat apples and oranges distinctly. The results on the unstructured model isolate the saliency rule in the absence of weight sparsity and the structured/search based readings bring back the cost of recovery and the budget allocation aspect.

6.4.1 Unstructured and Semi-Structured Methods

The unstructured pruning results allow three clean comparisons across increasing levels of local sophistication: magnitude baseline, Wanda (activation-aware saliency), and SparseGPT (second-order compensation).

The SparseGPT results primarily highlight the instability of pure magnitude pruning at high sparsity. At 50% sparsity on OPT-13B, magnitude pruning produces a perplexity of 1.2×10^4 compared to the dense model's 10.13. This indicates severe representational degradation under weight selection based solely on individual magnitude. SparseGPT reduces the same setting to 11.17, corresponding to a $+1.04$ increase over the dense model.

The comparison between Wanda and SparseGPT on LLaMA models is also instructive. At 50% sparsity on LLaMA-7B, Wanda achieves 7.26 versus SparseGPT's 7.22, a negligible 0.04 point difference. On LLaMA-13B, Wanda actually outperforms SparseGPT by 0.06 points (6.15 versus 6.21). This apparent paradox reflects two facts. First, the Wanda activation norm $\|x_j\|_2$ is an effective proxy for the weight's contribution to the output, particularly in models with large activation outliers, which LLaMA models exhibit. Second, the calibration set used for Hessian construction in SparseGPT introduces variance that at moderate sparsity levels can offset the theoretical advantage of curvature information. At higher sparsity levels (70%+), the second-order advantage of SparseGPT becomes more consistent, because local reconstruction errors accumulate and the precise compensation direction matters more.

6.4.2 Structured Methods

Both CoFi and ZipLM obtain similar accuracy at similar speedups but employ very different preparation pipeline. CoFi obtains 80.6 on MNLI with about $12\times$ speedups after 20 hours of combined mask and weight optimization, whereas ZipLM gets 81.7 at $35\times$ through scheduled fine-tuning interspersed with structured pruning stages and dedicated hardware benchmarking. Here the accuracies represent the accuracy of the whole pipeline not only the pruning criterion, because the recovery approaches they employ differ greatly and thus the two methods are not directly comparable.

6.4.3 Cross-Method Comparison

The tables above support two conclusions. At 50% unstructured sparsity on OPT-13B, magnitude pruning collapses to a perplexity of 1.2×10^4 while SparseGPT reaches 11.17 (Table 6.1), showing that second-order compensation is necessary once individual weight interactions become non-negligible. At 70% unstructured sparsity on Mistral-7B, EvoPress reaches 14.42 perplexity while uniform allocation reaches 23.08 (Table 6.3), and at 37.5% depth sparsity on Llama-2-7B, EvoPress gives 17.98 against ShortGPT's 70.94, showing that global allocation search becomes the dominant factor at high compression.

6.5 Limitations of the Reviewed Methods

The observations above suggest genuine differences between the methods. However, most of the disparity observed is can simply be a reflection of how each paper was evaluated. There are three consistent issues which have affected nearly every cross-paper assertion: unequal testing scenarios, disparate recovery budget sizes and sensitivity to a calibration set whose effect is usually not examined.

6.5.1 Evaluation Protocol Heterogeneity

A significant drawback in the pruning literature is the lack of a universal evaluation protocol. Approaches are presented under five, and possibly more, different evaluation schemes. The first reports language model perplexity under fixed nominal sparsity, while the second measures zero-shot task performance accuracy with or without recovery. The third instead reports supervised fine-tuning performance accuracy with or without recovery, whereas the fourth measures actual wall-clock speedup or latency given particular hardware. Finally, the fifth reports FLOPs reduction compared to the dense model.

These five schemes are incommensurable. A method presenting 99% dense accuracy after $2\times$ FLOPs reduction addresses a different evaluation goal compared to another method presenting $1.5\times$ wall-clock speedup at a $< 1\%$ accuracy drop. FLOPs is a hardware-agnostic theoretical metric that does not necessarily linearly translate to wall-clock runtime, which depends on memory bandwidth, batch size, and the particular accelerator execution model. Certain sparse execution formats require hardware support, while the simple unstructured masks often do not

result in a proportional wall-clock speedup over standard dense kernels.

The fragmentation of evaluation makes it difficult to compare across papers. If a table presents Wanda [Sun et al., 2024], CoFi [Xia et al., 2022], LLM-Pruner [Ma et al., 2023], ZipLM [Kurtić et al., 2023], and EvoPress [Sieberling et al., 2025] in the same rows, one implicitly assumes that they all aim to solve the same problem. In reality, each method targets different budgets, recovery schemes, and deployment targets. When summarized without qualification, the methods present a false picture of one single Pareto frontier.

6.5.2 Recovery Budget Inconsistency

Relatedly is the issue of different recovery budgets. It's expected that structured pruning results in a larger loss in accuracy than unstructured pruning for a given sparsity, making the recovery stage particularly important [Xu et al., 2026]. The difference in terminology of "post-training pruning" versus "pruning with fine-tuning" in reality represents a wide range of recovery costs that contribute to reported accuracy. Among the methods in the above literature, Fisher post-training pruning only tunings the masks with zero weight updates and total tuning time about 39 sec. LLM-Pruner tunings using 2.5 hrs of LoRA recovery on 50k examples with rank 8 adapter. ZipLM tunes over 8-10 epochs per pruning steps with the whole model fine-tuning and full distillation. CoFi joint tunes mask and weights in 40 epochs with a task specific teacher model.

An appropriate comparison needs to keep recovery budgets constant. The model of 40 epochs fine-tuning for CoFi should not be compared to LLM-Pruner's 2.5 hrs of LoRA recovery. Given the same budget, the gap would be much smaller. Very few articles performed such fair comparisons, and thus, some accuracy differences we see in the literature should be attributed to the amount of recovery budget.

6.5.3 Calibration Set Sensitivity

All post-training pruning techniques rely on a calibration set in order to compute either saliency scores, Hessian surrogates, or Fisher information matrices. The specific selection of the calibration set results in differences in the resulting pruned model which are usually not explored in a principled way.

For language models, it is a common practice to use 128–2048 examples from a general language corpus like C4 or WikiText. This is a small fraction of the model's training distribution, and may not be representative of activation statistics used to determine saliency. LLM-Pruner calibrates using 10 short examples, which is dramatically fewer than SparseGPT's 128 examples and may contribute to variance in its Taylor scores, and consequently in its reported zero-shot accuracies across trials. More generally, all methods employing a calibration set make the implicit assumption that a ranking based on calibration examples transfers to the whole evaluation distribution. It has recently been shown that the calibration set is extremely important in post-training pruning, especially for high sparsity rates, and that pruning works better with calibration data that matches the pre-training distribution [Bandari et al., 2024; Ji et al., 2025].ji2025calibration,bandari2024calibration.

6.6 Conclusion

The reviewed methods score different units, remove different structures, and require different recovery budgets, but they converge on the same limitation at high compression: local decisions are made before the joint effect of all removals is known. Magnitude and activation-aware methods score individual weights; gradient and second-order methods improve those scores but still apply them one unit at a time; dependency-aware structured methods such as LLM-Pruner rank groups independently across layers. The empirical results show the consequence directly: local ranking degrades rapidly at aggressive depth or sparsity budgets, and search over complete configurations remains stronger.

Chapter 7

Conclusion

This report has offered a survey of the state of the art methods in pruning Transformers. We've highlighted that while local scoring approaches may perform well for relatively low compression levels, they are limited at high sparsity levels as they do not take into account the interaction between components that were pruned. As the ratio of compression increases, how the sparsity budget is allocated across the layers becomes of utmost importance and local scoring approaches are unable to do so.

From the studies of the current methods, it can be observed that the Transformer pruning task needs to be viewed from a new perspective which is that the pruning should not be modeled as a ranking task in local, but rather as a configuration searching task in global in future work. This view can provide better insights into the dependencies of pruned parts with each other and lead to an efficient pruning strategy which achieves more significant compression ratio.

In summary, from the disadvantages of local scoring methods to prune Transformer, we can clearly see that when designing a pruning method for Transformer, we should not neglect the interaction and configuration of the Transformer globally. We believe that the future works in this area should be based on searching strategies pruning framework which can properly search the huge configuration space of pruning and attain optimum performance in high sparsity.

Bibliographie

Ba, J. L., Kiros, J. R., & Hinton, G. E. [2016]. Layer normalization. *arXiv preprint arXiv:1607.06450*. https://doi.org/10.48550/arXiv.1607.06450

Bai, G., Chai, Z., Ling, C., Wang, S., Lu, J., Zhang, N., Chen, L., Tian, D., Su, M., Wang, H., & Zhao, L. [2024]. SparseLLM: Towards global pruning of pre-trained language models. *Advances in Neural Information Processing Systems*, *37*.

Bandari, A., Yin, L., Hsieh, C.-Y., Jaiswal, A. K., Chen, T., Shen, L., Krishna, R., & Liu, S. [2024]. Is c4 dataset optimal for pruning? an investigation of calibration data for LLM pruning. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen [Eds.], *Proceedings of the 2024 conference on empirical methods in natural language processing* [pp. 18089–18099]. Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.1004

Beltagy, I., Peters, M. E., & Cohan, A. [2020]. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*. https://doi.org/10.48550/arXiv.2004.05150

Bengio, Y., Léonard, N., & Courville, A. C. [2013]. Estimating or propagating gradients through stochastic neurons for conditional computation. *CoRR*, *abs/1308.3432*. http://arxiv.org/abs/1308.3432

Bottou, L., Curtis, F. E., & Nocedal, J. [2018]. Optimization methods for large-scale machine learning. *SIAM Review*, *60*[2], 223–311. https://doi.org/10.1137/16M1080173

Cheng, H., Zhang, M., & Shi, J. Q. [2024a]. A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*[12], 10558–10578. https://doi.org/10.1109/TPAMI.2024.3447085

Cheng, H., Zhang, M., & Shi, J. Q. [2024b]. A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*[12], 10558–10578. https://doi.org/10.1109/TPAMI.2024.3447085

Clevert, D.-A., Unterthiner, T., & Hochreiter, S. [2016]. Fast and accurate deep network learning by exponential linear units (ELUs) [Poster]. *International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.1511.07289

Cover, T. M., & Thomas, J. A. [2006]. *Elements of information theory* [2nd ed.]. John Wiley & Sons. https://doi.org/10.1002/047174882X

Denil, M., Shakibi, B., Dinh, L., Ranzato, M., & de Freitas, N. [2013]. Predicting parameters in deep learning. *Advances in Neural Information Processing Systems*, *26*, 2148–2156.

https://proceedings.neurips.cc/paper/2013/hash/7fec306d1e665bc9c748b5d2b99a6e97-Abstract.html

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. [2019]. BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, & T. Solorio [Eds.], *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: Human language technologies, volume 1 (long and short papers)* [pp. 4171–4186]. Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Dong, X., Chen, S., & Pan, S. J. [2017]. Learning to prune deep neural networks via layer-wise optimal brain surgeon. *Advances in Neural Information Processing Systems*, 4857–4867. https://proceedings.neurips.cc/paper/2017/hash/c5dc3e08849bec07e33ca353de62ea04-Abstract.html

Frankle, J., & Carbin, M. [2019]. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *International Conference on Learning Representations*. https://openreview.net/forum?id=rJl-b3RcF7

Frantar, E., & Alistarh, D. [2023]. SparseGPT: Massive language models can be accurately pruned in one-shot. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett [Eds.], *Proceedings of the 40th international conference on machine learning* [pp. 10323–10337, Vol. 202]. PMLR. https://proceedings.mlr.press/v202/frantar23a.html

Frantar, E., Ashkboos, S., Hoefler, T., & Alistarh, D. [2023]. OPTQ: Accurate post-training quantization for generative pre-trained transformers [The arXiv version is titled GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers]. *International Conference on Learning Representations*. https://openreview.net/forum?id=tcbBPnfwxS

Gale, T., Elsen, E., & Hooker, S. [2019]. The state of sparsity in deep neural networks. *arXiv preprint arXiv:1902.09574*.

Gale, T., Zaharia, M., Young, C., & Elsen, E. [2020]. Sparse gpu kernels for deep learning. *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1–14. https://doi.org/10.1109/SC41405.2020.00021

Glorot, X., & Bengio, Y. [2010]. Understanding the difficulty of training deep feedforward neural networks. In Y. W. Teh & M. Titterington [Eds.], *Proceedings of the thirteenth international conference on artificial intelligence and statistics* [pp. 249–256, Vol. 9]. PMLR. https://proceedings.mlr.press/v9/glorot10a.html

Goodfellow, I., Bengio, Y., & Courville, A. [2016]. *Deep learning*. MIT Press. https://doi.org/10.5555/3086952

Guo, Y., Yao, A., & Chen, Y. [2016]. Dynamic network surgery for efficient dnns. *Advances in Neural Information Processing Systems*, *29*, 1379–1387. https://papers.nips.cc/paper_files/paper/2016/hash/2823f4797102ce1a1aec05359cc16dd9-Abstract.html

Han, S., Mao, H., & Dally, W. J. [2016]. Deep compression: Compressing deep neural networks with pruning, trained quantization and huffman coding. *4th International Conference on Learning Representations*. https://arxiv.org/abs/1510.00149

Han, S., Pool, J., Tran, J., & Dally, W. [2015]. Learning both weights and connections for efficient neural network. *Advances in Neural Information Processing*

Systems, *28*, 1135–1143. https : / / proceedings . neurips . cc / paper / 2015 / hash / ae0eb3eed39d2bcef4622b2499a05fe6-Abstract.html

Hassibi, B., & Stork, D. G. [1993]. Second order derivatives for network pruning: Optimal brain surgeon. In S. J. Hanson, J. D. Cowan, & C. L. Giles [Eds.], *Advances in neural information processing systems* [pp. 164–171, Vol. 5]. Morgan Kaufmann. https://proceedings.neurips.cc/paper_files/paper/1992/file/303ed4c69846ab36c2904d3ba8573050-Paper.pdf

He, K., Zhang, X., Ren, S., & Sun, J. [2016]. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

Hendrycks, D., & Gimpel, K. [2016]. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*. https://doi.org/10.48550/arXiv.1606.08415

Hoefler, T., Alistarh, D., Ben-Nun, T., Dryden, N., & Peste, A. [2021]. Sparsity in deep learning: Pruning and growth for efficient inference and training in neural networks. *Journal of Machine Learning Research*, *22*[241], 1–124. http://jmlr.org/papers/v22/21-0366.html

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., … Sifre, L. [2022]. Training compute-optimal large language models. *Advances in Neural Information Processing Systems*, *35*. https://doi.org/10.48550/arXiv.2203.15556

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. [2022]. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.2106.09685

Hu, Y., Zhao, K., Huang, W., Chen, J., & Zhu, J. [2024]. Accelerating transformer pre-training with 2:4 sparsity [21–27 Jul]. In R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, & F. Berkenkamp [Eds.], *Proceedings of the 41st international conference on machine learning* [pp. 19531–19543, Vol. 235]. PMLR. https://proceedings.mlr.press/v235/hu24r.html

Huang, W., Hu, Y., Jian, G., Zhu, J., & Chen, J. [2025]. Pruning large language models with semi-structural adaptive sparse training. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*[23], 24167–24175. https://doi.org/10.1609/aaai.v39i23.34592

Janusz, M., Wojnar, T., Li, Y., Benini, L., & Adamczewski, K. [2025]. One shot vs. iterative: Rethinking pruning strategies for model compression. *ECAI 2025: 28th European Conference on Artificial Intelligence*, *413*, 2514–2521. https://doi.org/10.3233/FAIA251100

Ji, Y., Xiang, Y., Li, J., Xia, Q., Li, P., Duan, X., Wang, Z., & Zhang, M. [2025]. Beware of calibration data for pruning large language models. *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=x83w6yGIWb

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. [2020]. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*. https://doi.org/10.48550/arXiv.2001.08361

Kingma, D. P., & Ba, J. [2015]. Adam: A method for stochastic optimization. *International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.1412.6980

Kurtic, E., Campos, D., Nguyen, T., Frantar, E., Kurtz, M., Fineran, B., Goin, M., & Alistarh, D. [2022]. The optimal BERT surgeon: Scalable and accurate second-order pruning for large language models. In Y. Goldberg, Z. Kozareva, & Y. Zhang [Eds.], *Proceedings of the 2022 conference on empirical methods in natural language processing* [pp. 4163–4181]. Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.emnlp-main.279

Kurtić, E., Frantar, E., & Alistarh, D. [2023]. ZipLM: Inference-aware structured pruning of language models. *Advances in Neural Information Processing Systems*, *36*, 65597–65617. https://proceedings.neurips.cc/paper_files/paper/2023/hash/ced46a50befedcb884ccf0cbe8c3ad23-Abstract-Conference.html

Kwon, W., Kim, S., Mahoney, M. W., Hassoun, J., Keutzer, K., & Gholami, A. [2022]. A fast post-training pruning framework for transformers. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh [Eds.], *Advances in neural information processing systems* [pp. 24101–24116, Vol. 35]. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2022/file/987bed997ab668f91c822a09bce3ea12-Paper-Conference.pdf

LeCun, Y., Denker, J. S., & Solla, S. A. [1990]. Optimal brain damage [NIPS 1989]. In D. S. Touretzky [Ed.], *Advances in neural information processing systems* [pp. 598–605, Vol. 2]. Morgan Kaufmann. https://proceedings.neurips.cc/paper_files/paper/1989/hash/6c9882bbac1c7093bd25041881277658-Abstract.html

Li, H., Kadav, A., Durdanovic, I., Samet, H., & Graf, H. P. [2017]. Pruning filters for efficient convnets. *International Conference on Learning Representations*. https://openreview.net/forum?id=rJqFGTslg

Li, L., Dong, P., Tang, Z., Liu, X., Wang, Q., Luo, W., Xue, W., Liu, Q., Chu, X., & Guo, Y. [2024]. Discovering sparsity allocation for layer-wise pruning of large language models. *Advances in Neural Information Processing Systems*, *37*, 141292–141317. https://doi.org/10.52202/079017-4487

Liu, Z., Li, J., Shen, Z., Huang, G., Yan, S., & Zhang, C. [2017]. Learning efficient convolutional networks through network slimming. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2755–2763. https://doi.org/10.1109/ICCV.2017.298

Loshchilov, I., & Hutter, F. [2019]. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.1711.05101

Louizos, C., Welling, M., & Kingma, D. P. [2018]. Learning sparse neural networks through L_0 regularization. *International Conference on Learning Representations*. https://openreview.net/forum?id=H1Y8hhg0b

Ma, X., Fang, G., & Wang, X. [2023]. LLM-Pruner: On the structural pruning of large language models. *Advances in Neural Information Processing Systems*, *36*, 21702–21720. https://proceedings.neurips.cc/paper_files/paper/2023/hash/44956951349095f74492a5471128a7e0-Abstract-Conference.html

Maas, A. L., Hannun, A. Y., & Ng, A. Y. [2013]. Rectifier nonlinearities improve neural network acoustic models [WDLASL 2013]. *Proceedings of the ICML Workshop on Deep Learning for Audio, Speech and Language Processing*. https://ai.stanford.edu/~amaas/papers/relu_hybrid_icml2013_final.pdf

Mallya, A., & Lazebnik, S. [2018]. Packnet: Adding multiple tasks to a single network by iterative pruning. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 7765–7773. https://openaccess.thecvf.com/content_cvpr_2018/html/Mallya_PackNet_Adding_Multiple_CVPR_2018_paper.html

McCulloch, W. S., & Pitts, W. [1943]. A logical calculus of the ideas immanent in nervous activity. *The Bulletin of Mathematical Biophysics*, *5*[4], 115–133. https://doi.org/10.1007/BF02478259

Michel, P., Levy, O., & Neubig, G. [2019]. Are sixteen heads really better than one? In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, & R. Garnett [Eds.], *Advances in neural information processing systems* [pp. 14014–14024, Vol. 32]. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2019/file/2c601ad9d2ff9bc8b282670cdd54f69f-Paper.pdf

Mishra, A., Latorre, J. A., Pool, J., Stosic, D., Stosic, D., Venkatesh, G., Yu, C., & Micikevicius, P. [2021]. Accelerating sparse deep neural networks. *arXiv preprint arXiv:2104.08378*. https://doi.org/10.48550/arXiv.2104.08378

Molchanov, D., Ashukha, A., & Vetrov, D. [2017]. Variational dropout sparsifies deep neural networks. In D. Precup & Y. W. Teh [Eds.], *Proceedings of the 34th international conference on machine learning* [pp. 2498–2507, Vol. 70]. PMLR. https://proceedings.mlr.press/v70/molchanov17a.html

Nair, V., & Hinton, G. E. [2010]. Rectified linear units improve restricted boltzmann machines. *Proceedings of the 27th International Conference on Machine Learning*, 807–814. https://www.cs.toronto.edu/~hinton/absps/reluICML.pdf

NVIDIA. [2026]. *Matrix multiplication background user's guide*. Retrieved May 24, 2026, from https://docs.nvidia.com/deeplearning/performance/dl-performance-matrix-multiplication/index.html

NVIDIA Corporation. [2020]. *NVIDIA A100 Tensor Core GPU Architecture* [White paper] [Accessed: 2026-06-01]. NVIDIA Corporation. https://images.nvidia.com/aem-dam/en-zz/Solutions/data-center/nvidia-ampere-architecture-whitepaper.pdf

OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. [2023]. GPT-4 technical report. *CoRR*, *abs/2303.08774*. https://doi.org/10.48550/arXiv.2303.08774

PyTorch Contributors. [2026a]. *torch.numel*. Retrieved May 24, 2026, from https://docs.pytorch.org/docs/stable/generated/torch.numel.html

PyTorch Contributors. [2026b]. *torch.Tensor.element_size*. Retrieved May 24, 2026, from https://docs.pytorch.org/docs/stable/generated/torch.Tensor.element_size.html

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. [2019]. *Language models are unsupervised multitask learners* [tech. rep.]. OpenAI. https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. [1986]. Learning representations by back-propagating errors. *Nature*, *323*[6088], 533–536. https://doi.org/10.1038/323533a0

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. [2019]. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *Proceedings of the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (EMC2) at NeurIPS 2019*. https://doi.org/10.48550/arXiv.1910.01108

Sanh, V., Wolf, T., & Rush, A. [2020]. Movement pruning: Adaptive sparsity by fine-tuning. *Advances in Neural Information Processing Systems*, *33*, 20378–20389. https://proceedings.neurips.cc/paper/2020/hash/eae15aabaa768ae4a5993a8a4f4fa6e4-Abstract.html

Sieberling, O., Kuznedelev, D., Kurtic, E., & Alistarh, D. [2025]. EvoPress: Accurate dynamic model compression via evolutionary search [13–19 Jul]. *Proceedings of the 42nd International Conference on Machine Learning*, *267*, 55556–55590. https://proceedings.mlr.press/v267/sieberling25a.html

Sun, M., Liu, Z., Bair, A., & Kolter, J. Z. [2024]. A simple and effective pruning approach for large language models. *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=PxoFut3dWW

Sze, V., Chen, Y.-H., Yang, T.-J., & Emer, J. S. [2017]. Efficient processing of deep neural networks: A tutorial and survey. *Proceedings of the IEEE*, *105*[12], 2295–2329. https://doi.org/10.1109/JPROC.2017.2761740

Tang, S., Sieberling, O., Kurtic, E., Shen, Z., & Alistarh, D. [2025]. DarwinLM: Evolutionary structured pruning of large language models. *CoRR*, *abs/2502.07780*. https://doi.org/10.48550/arXiv.2502.07780

Tay, Y., Dehghani, M., Bahri, D., & Metzler, D. [2022]. Efficient transformers: A survey. *ACM Computing Surveys*, *55*[6], 1–28. https://doi.org/10.1145/3530811

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. [2023]. Llama 2: Open foundation and fine-tuned chat models. *CoRR*, *abs/2307.09288*. https://doi.org/10.48550/arXiv.2307.09288

Vapnik, V. N. [1998]. *Statistical learning theory*. Wiley-Interscience. https://www.wiley.com/en-us/Statistical%2BLearning%2BTheory-p-9780471030034

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. [2017]. Attention is all you need. *Advances in Neural Information Processing Systems*, *30*, 5998–6008. https://doi.org/10.5555/3295222.3295349

Voita, E., Talbot, D., Moiseev, F., Sennrich, R., & Titov, I. [2019]. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. In A. Korhonen, D. Traum, & L. Màrquez [Eds.], *Proceedings of the 57th annual meeting of the association for computational linguistics* [pp. 5797–5808]. Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1580

Wei, X., Zhang, Y., Zhang, X., Gong, R., Zhang, S., Zhang, Q., Yu, F., & Liu, X. [2022]. Outlier suppression: Pushing the limit of low-bit transformer language models. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh [Eds.], *Advances in neural information processing systems* [pp. 17402–17414, Vol. 35]. Neural Information Processing Systems Foundation. https://doi.org/10.52202/068431-1265

Xia, M., Zhong, Z., & Chen, D. [2022]. Structured pruning learns compact and accurate models. In S. Muresan, P. Nakov, & A. Villavicencio [Eds.], *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)* [pp. 1513–1528]. Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.107

Xu, Z., Xu, Y., Xu, H., Liao, Y., Yao, Z., & Xie, Z. [2026]. Lightweight and post-training structured pruning for on-device large language models. *IEEE Transactions on Mobile Computing*, *25*[4], 5377–5392. https://doi.org/10.1109/TMC.2025.3629756

Zhu, M., & Gupta, S. [2018]. To prune, or not to prune: Exploring the efficacy of pruning for model compression. *International Conference on Learning Representations Workshop*. https://arxiv.org/abs/1710.01878

Webographie

Cai, J. [2024]. Accelerating BERT with semi-structured (2:4) sparsity [Created: 2024-04-22; last updated: 2025-09-29; accessed: 2026-06-01]. https://docs.pytorch.org/tutorials/advanced/semi_structured_sparse.html